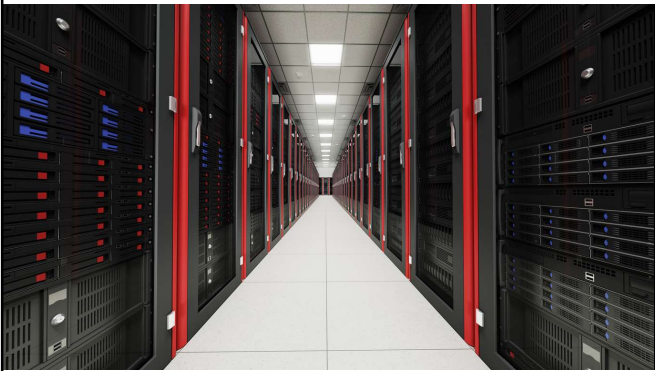
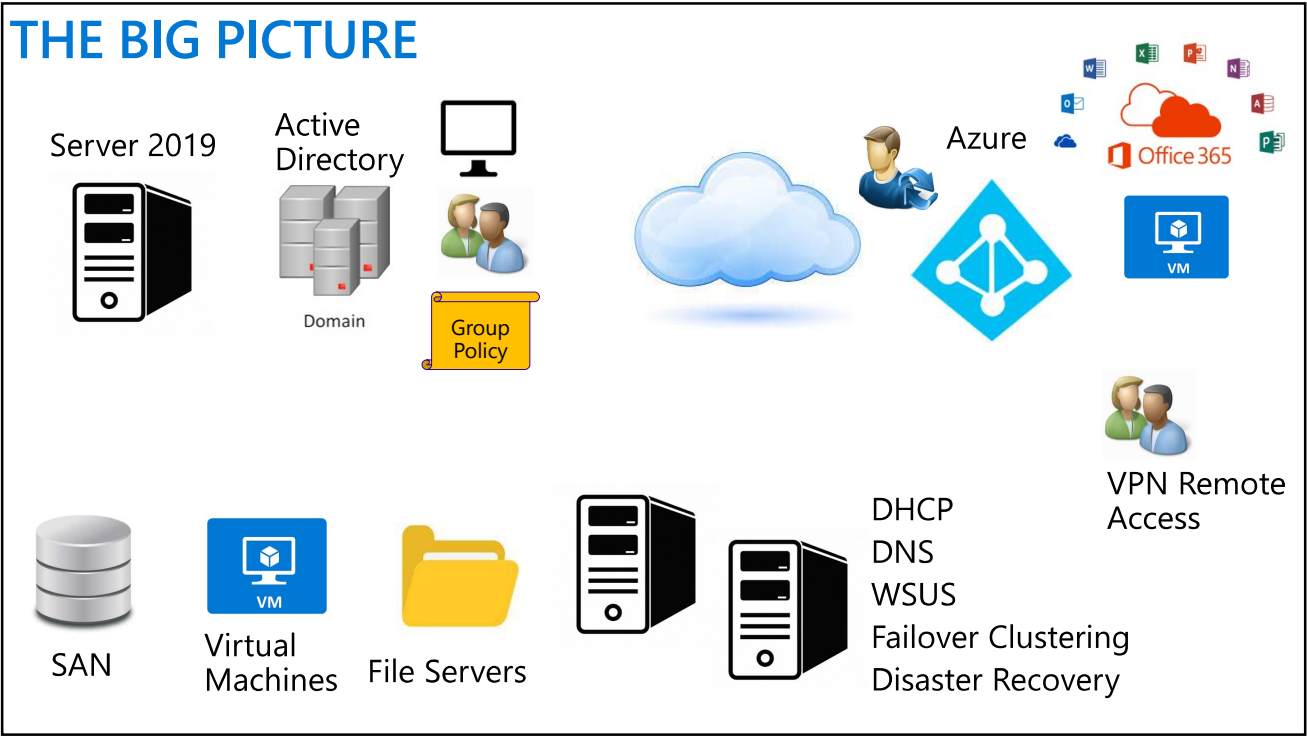




Windows Server 2019 Administration

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Azure Security Engineer,
Azure Solutions Architect,
Azure Administrator,
MCSE,
CISSP

THE BIG PICTURE



Who is this course for?

1. Introducing Windows Server 2019
2. Active Directory Domain Services on Windows Server 2019
3. Azure Active Directory (AzureAD)
4. Managing DHCP and DNS
5. Managing File Servers and Storage
6. Hyper-V Virtualization
7. Disaster Recovery in Windows Server 2019
8. Managing SAN Storage and Failover Clustering
9. Implementing WSUS on Windows Server 2019
10. Implementing Remote Access on Windows Server 2019
11. Performance Monitoring in Windows Server 2019

Course outline

1. Introducing Windows Server 2019
2. Active Directory Domain Services on Windows Server 2019
3. Azure Active Directory (AzureAD)
4. Understanding IPv4 and IPv6
5. Managing DHCP and DNS
6. Managing File Servers and Storage
7. Hyper-V Virtualization
8. Disaster Recovery in Windows Server 2019
9. Managing SAN Storage and Failover Clustering
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12. Performance Monitoring in Windows Server 2019

Lab Setup

<https://RTSnetworking.com/demo>

Hyper-V lab

Windows Server 2019 ISO download

Enabling Hyper-V

Installing Windows Server 2019 using Hyper-V

Hyper-V post installation tasks

VirtualBox lab

Windows Server 2019 ISO download

Installing VirtualBox

Installing Windows Server 2019 using VirtualBox

VirtualBox post installation tasks



Module 1:
Introducing
Windows Server 2019

Module overview

This module introduces you to Windows Server 2019

Lessons

- Requirements for Windows Server 2019
- Windows Server 2019 Editions
- Deployment Options
- Overview of Windows Server Core
- Overview of administration methods and tools

Lesson 1 overview

In this lesson, you'll learn about Windows Server 2019 editions and their capabilities

- Topics
 - Windows Server 2019 editions
 - Hardware requirements for Windows Server 2019
 - Overview of deployment options
 - Servicing channels for Windows Server
 - Licensing and activation for Windows Server

Windows Server 2019 editions

- Windows Server 2019 is released in four editions:
 - Windows Server 2019 Essential
 - Windows Server 2019 Standard
 - Windows Server 2019 Datacenter
 - Hyper-V server 2019

Edition Comparison

- <https://docs.microsoft.com/en-us/windows-server/get-started-19/editions-comparison-19>

Hardware requirements for Windows Server 2019

- Hardware requirements will vary depending on:
 - Server roles
 - Many roles have specific requirements
 - Resource usage

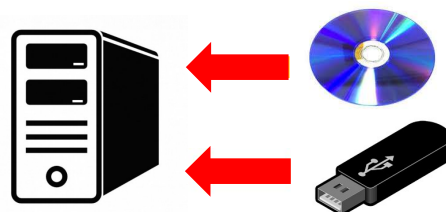
Minimum hardware requirements for Windows Server 2019:

Component	Requirement
Processor architecture	64 bit
Processor speed	1.4 gigahertz (GHz)
RAM	512 MB
Hard drive space	32 GB

Overview of deployment options (1 of 2)

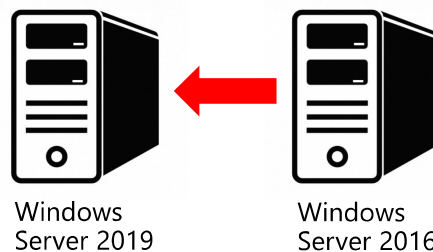
Clean install:

- Boot the machine or VM from the Windows Server 2019 media
- Choose the installation language, time and currency formats, and keyboard layout
- Choose the architecture (either Standard or Datacenter) with or without Desktop Experience
- Accept the license
- Choose custom installation
- Choose the volume that will host the installation



Overview of deployment options (2 of 2)

- In-place upgrade
 - Insert the disk or mount the ISO of Windows Server 2019 media and then run **Setup.exe**
 - Choose the architecture (either **Standard** or **Datacenter**) with or without **Desktop Experience**
 - Accept the license
 - Choose what to keep: personal files and apps, or nothing



Demo: Windows Server 2019 Installation

Prerequisite: Create a new virtual machine

- ☐ Name = RTS-DC1
- ☐ Generation =Generation 2
- ☐ Memory = 2048 MB (2GB) recommended 4096 MB (4GB)
- ☐ Virtual Switch = Name: vSwitch, Type: External
- ☐ VHD: Accept default
- ☐ ISO: Browse to downloaded Windows Server 2019 ISO file

VirtualBox Demo: <https://RTSnetworking.com/demo>

Servicing channels for Windows Server

You can use servicing channels to choose whether new features and functionality will be delivered regularly during a server's production lifespan, or when to move to a new server version

- There are two release channels:
 - Long-Term Servicing Channel
 - A new major version of Windows Server is released every 2-3 years. Users are entitled to 5 years of mainstream support and 5 years of extended support. This channel is appropriate for systems that require a longer servicing option and functional stability. The Long-Term Servicing Channel will continue to receive security and non-security updates, but it will not receive the new features and functionality.
 - Semi-Annual Channel
 - The Semi-Annual Channel is perfect for customers who are innovating quickly to take advantage of new operating system capabilities at a faster pace, focused in on containers and microservices. Windows Server products in the Semi-Annual Channel will have new releases available twice a year, in spring and fall. Each release in this channel will be supported for 18 months from the initial release.

So what is Windows Server 20H2 (H2- Second Half)?

Semi-Annual Channel

Version	Servicing option	Availability date
20H2	Semi-Annual Channel	2020-10-20
2004	Semi-Annual Channel	2020-05-27
1909	Semi-Annual Channel	2019-11-12
1809	Semi-Annual Channel	2019-03-28
1809	Semi-Annual Channel (Targeted)	2018-11-13
1803	Semi-Annual Channel	2018-07-10

Licensing and activation models for Windows Server (1 of 2)

Licensing for Windows Server **Standard** and **Datacenter** is based on the number of cores, not processors

- Each Windows Server has the following minimum license requirement:
 - All physical cores must be licensed
 - There must be 8 core licenses per processor
 - There must be 16 core licenses per server
- Client access licenses are required for each user or device that connects to the server for any purpose

Licensing and activation models for Windows Server (2 of 2)

To ensure that your organization has the proper licenses, you must activate every copy of Windows Server that you install

- Windows Server activation methods:
 - Manual activation requires a product key
 - Automatic activation options:
 - Key Management Services
 - Active Directory-based activation
 - Multiple Activation Key
 - Automatic virtual machine activation

Demo: Software Licensing Manager

Software Licensing Manager script

- SLMGR.VBS is a built-in script to manage activation
- Windows Vista and newer client operating systems
- Windows Server 2008 and newer server operating systems

Lesson 2 overview

In this lesson, you'll learn about the differences between Server Core and Windows Server with Desktop Experience, and when one is the preferred option

- Topics
 - Server Core vs. Windows Server with Desktop Experience
 - Server Core installation and post-installation tasks
 - Install features on demand
 - Use sconfig in Server Core

Server Core vs. Windows Server with Desktop Experience

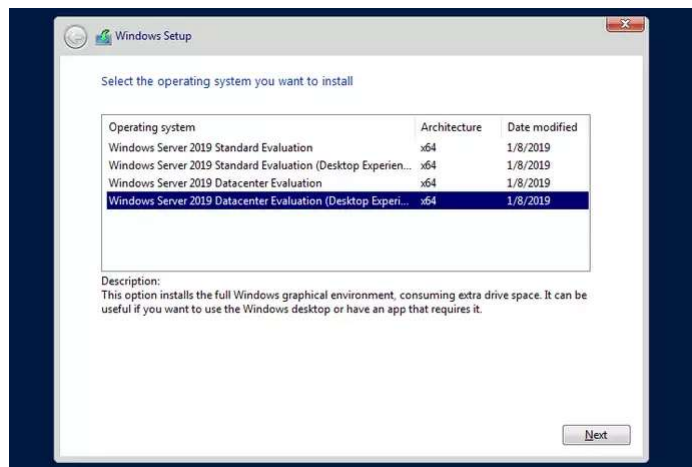
The following table lists the major advantages and disadvantages of Server Core

Advantages	Disadvantages
Small footprint that uses fewer server resources and less disk space, as little as 6 GB for a basic installation	Some applications are not supported on a Server Core installation.
Because Server Core installs fewer components, there are fewer software updates. This reduces the number of monthly restarts required and the time required for you to service Server Core.	Some roles and role services are not available.
The small attack surface makes Server Core much less vulnerable to exploits.	No local Graphical User Interface (GUI)

Server Core installation and post-installation tasks

To install Server Core:

- Connect to the installation source
- Choose:
 - Language
 - Time and currency
 - Keyboard
- Select the operating system to install
- Accept license
- Choose installation type
 - Upgrade
 - Custom
- Choose install disk
- Provide admin password



Using sconfig in Server Core

- sconfig is a menu-based utility that allows you configure Server Core
- sconfig eliminates the need for scripting configuration settings

```
Server Configuration

1) Domain/Workgroup:          Domain: Contoso.com
2) Computer Name:             SEA-SVR4
3) Add Local Administrator
4) Configure Remote Management Enabled
5) Windows Update Settings:   Manual
6) Download and Install Updates
7) Remote Desktop:           Disabled
8) Network Settings
9) Date and Time
10) Telemetry settings        Unknown
11) Windows Activation

12) Log Off User
13) Restart Server
14) Shut Down Server
15) Exit to Command Line

Enter number to select an option: 
```

Lesson 3 overview

In this lesson, you'll learn about Windows Server administration best practices and the tools used for management.

- Topics
 - Overview of the least-privilege administration concept
 - Delegated privileges
 - Jump servers
 - Overview of the Windows Server Admin Center
 - Server Manager
 - Remote Server Administration Tools (RSAT)
 - Introduction to Windows PowerShell

Overview of the least-privilege administration concept

Most security breaches or data loss incidents are the result of human error, malicious activity, or a combination of both. Least privilege is the concept of restricting access rights for users and computing processes to only those resources absolutely required to perform their job roles.

The principle states that all users should log on with a user account that has the absolute minimum permissions necessary to complete the current task and nothing more. Doing so provides protection against malicious code, among other attacks. This principle applies to computers and the users of those computers.



Day to day
standard user
account for IT
admins



Full admin account only
used to perform
administration
functions



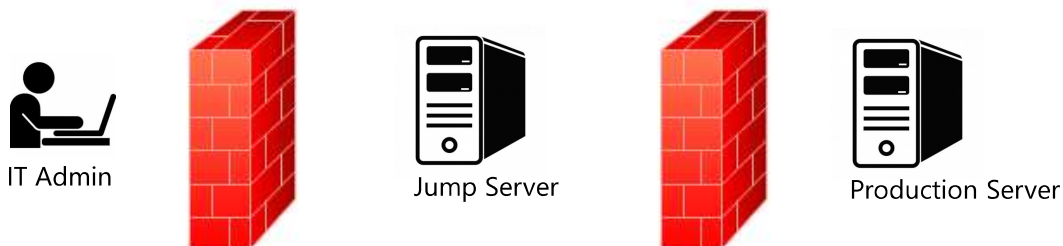
Domain
Controller

Delegated privileges

- Accounts that are members of high-privilege groups such as **Enterprise Admins** and **Domain Admins** need to be guarded, but occasionally non-admins need rights to perform certain functions, such as resetting passwords or modifying group memberships.
- Built-in groups with pre-defined admin rights exist to allow users to perform specific admin tasks. If those groups do not suit your needs, you can delegate more granular permissions by using the **Delegation of Control Wizard**.
 - The wizard has pre-defined tasks that can be assigned to users or groups, or custom permissions can be assigned.

Jump servers

- A jump server is a hardened server used to access and manage devices in a different security zone, such as between an internal network and a perimeter network



Overview of Windows Admin Center

- Windows Admin Center consolidates multiple admin tools into a single console that can be easily deployed and accessed through a web interface
- Windows Admin Center is a modular web application comprised of the following four modules:
 - Server manager
 - Failover clusters
 - Hyper-converged clusters
 - Windows 10 clients

Server Manager

- **Server Manager** allows server administrators to:
 - Manage the local server and remotely manage multiple servers
 - Configure the local server
 - Query event logs
 - Monitor status of services
 - Perform best practice analysis
 - Check performance monitors
- **Server Manager** initially opens to the dashboard, which provides quick access to:
 - Add roles and features
 - Add other servers to manage
 - Create a server group
 - Connect this server to cloud services

Remote Server Administration Tools

- To enable IT administrators to remotely manage roles and features in Windows Server from a computer that is running Windows 10 or Windows 8.1, use **RSAT**
- **RSAT** include:
 - Active Directory Domain Services tools
 - DHCP server tools
 - DNS server tools
 - File services tools
 - Group Policy management tools

<https://www.microsoft.com/download/details.aspx?id=45520>

Windows PowerShell

- Windows PowerShell is a command line shell and scripting language
- Windows PowerShell cmdlets execute in a Windows PowerShell console or can be executed as PowerShell scripts
- Cmdlets:
 - Are small commands that perform specific functions
- Modules:
 - Cmdlets specific to a product are packaged together and installed as modules
 - Some are installed with the product and some need to be added manually

Windows PowerShell

- PowerShell Console
 - Run PowerShell commands and execute scripts
- PowerShell ISE
 - PowerShell Integrated Scripting Environment (ISE) is a graphical user interface-based tool that allows you to:
 - Run commands, create, modify and execute scripts
- Windows PowerShell remote management:
 - Allows Windows PowerShell to remotely run cmdlets on other Windows systems

Windows PowerShell vs CMD Prompt

CMD Prompt

Ping

Ping rtsnetworking.com

Ping -t rtsnetworking.com

IPconfig

Ipconfig /all

DiskPart utility

Unique syntax only used in DiskPart

PowerShell

Verb-Noun

Test-connection rtsnetworking.com

Test-connection rtsnetworking.com -count 10

Get-eventlog

Get-eventlog -LogName System

PowerShell Help

Get-help

Get-help *event*

Get-help *eventlog*

Get-help get-eventlog

Get-help get-eventlog -detailed

Get-help get-eventlog -examples

Get-help get-eventlog -full

Get-help get-eventlog -online



Module 2: Active Directory Domain Services on Windows Server 2019

Module overview

This module introduces you to Windows Server 2019

Lessons

- Overview of AD DS
- Deploying domain controllers on Windows Server 2019
- Implementing Group Policy

Lesson 1

Topics:

- ☐ Understanding workgroups and domains
- ☐ Active Directory terms
- ☐ Active Directory Domains and Forests
- ☐ Understanding Active Directory objects
- ☐ Understanding Domain Controllers
- ☐ Global Catalog

Types of Networks

WORKGROUP

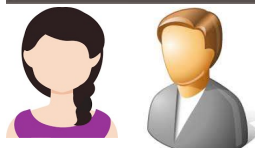
- No Centralized Authentication
- No Centralized Administration
- Max of 20 computers supported
- Low Security

DOMAIN

- Centralized Authentication
- Centralized Administration
- Unlimited number of computers
- High Security

Domain vs Workgroup (Non-Technical way to think)

Domain



Parents

*House Rules set by
parents*



Children

Workgroup



*Each person sets their
own rules*



Children

Active Directory Domain Services (AD DS)Terms

AD DS is composed of both logical and physical components

Logical components	Physical components
• Domains • Domain trees • Forests • OUs • Containers	• Domain Controllers • Read-only Domain Controllers

AD DS Domains and Forests

- A domain:
 - A repository for User, Computers and other objects
 - A replication boundary
 - An administrative boundary
- A domain controller is a server that has Active Directory Domain Services (AD DS) installed

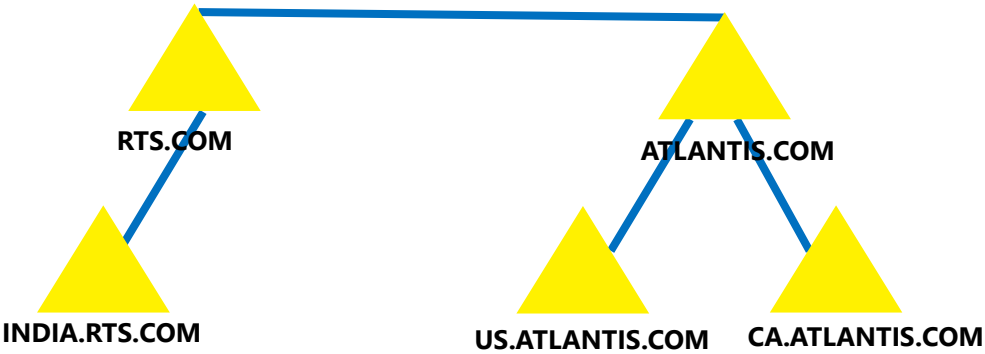
AD DS Domains and Forests



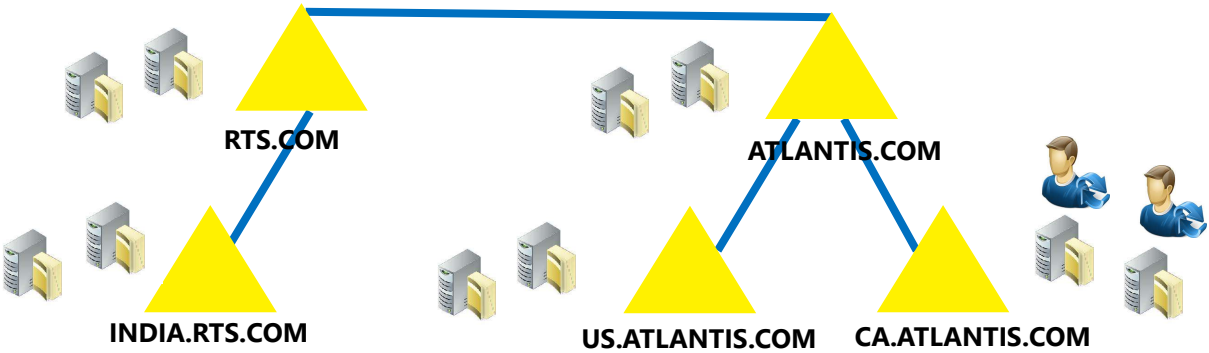
AD DS Domains and Forests

- A forest:
 - Is a security boundary
 - One or more domains that share a trust relationship
- Trust relationships:
 - A relationship between domains that allows access to resources in other domains within the same forest

AD DS Domains and Forests



AD DS Forest



RTSNetworking.com



RTS-Core (Server 2019)

IP Address: 192.168.1.252
Subnet Mask: 255.255.255.0
Default Gateway: (Blank)
Preferred DNS: 192.168.1.250
Alternate DNS: (Blank)



RTS-DC1 Domain Controller/ DNS Server (Server 2019)

IP Address: 192.168.1.250
Subnet Mask: 255.255.255.0
Default Gateway: (Blank)
Preferred DNS: 192.168.1.250
Alternate DNS: (Blank)



RTS-SVR1 (Server 2019)

IP Address: 192.168.1.251
Subnet Mask: 255.255.255.0
Default Gateway: (Blank)
Preferred DNS: 192.168.1.250
Alternate DNS: (Blank)

AD DS objects

- User objects
 - Authentication of the user at logon
 - Access control
- Group objects
 - Simplify assigning permissions
- Computer objects
 - Authentication of the computer at startup



Organizational Units and Containers

- Use containers to group objects within a domain:
 - You cannot apply GPOs to containers
 - Containers are used for system objects and as the default location for new objects

- Create OUs to:
 - Configure objects by assigning GPOs to them
 - Delegate administrative permissions



Container



Organizational
Unit

Domain Controllers

Domain controllers:

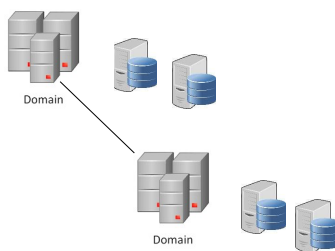
- Are servers that host the AD DS database (**Ntds.dit**) and **SYSVOL**
- Host the Kerberos authentication service and KDC services to perform authentication
- Have best practices for:
 - Availability:
 - Use at least two domain controllers in a domain



Domain

What is the global catalog?

- The global catalog:
 - Hosts a partial attribute set for other domains in the forest
 - Supports queries for objects throughout the forest
- In a single domain, you should configure all the domain controllers to hold a copy of the global catalog
- When you have multiple sites, you should also make at least one domain controller at each site a global catalog server



Lesson 2

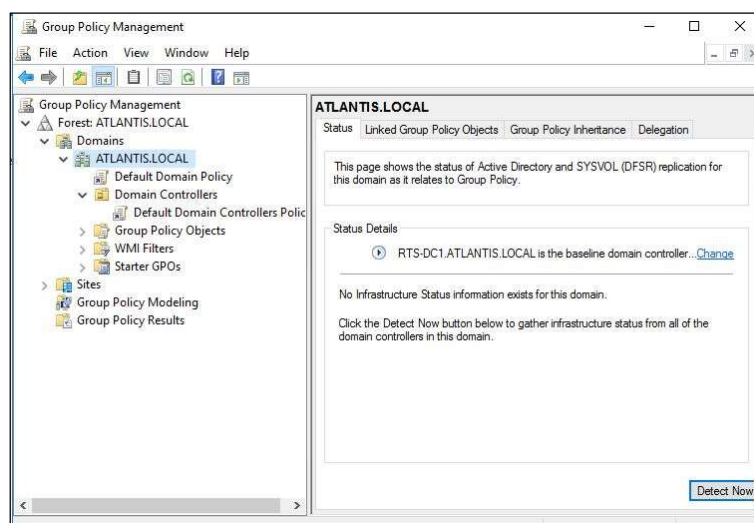
Topics:

- ☐ What are Group Policy Objects?
- ☐ Domain-based Group Policy Objects
- ☐ Group Policy Preferences
- ☐ Inheritance
- ☐ Group Policy Processing
- ☐ Gpupdate and Gpresult

What are GPOs?

- Group Policy is a powerful administrative tool
- You can use it to enforce various types of settings to a large number of users and computers
- Typically, you use GPOs to:
 - Apply security settings
 - Manage desktop application settings
 - Deploy application software
 - Manage Folder Redirection
 - Configure network settings

What are domain-based GPOs?



What are Group Policy Preferences?

Group Policy Preferences are a collection of Group Policy client-side extensions that deliver preference settings to domain-joined computers.

Preference settings differ from policy settings because users have a choice to alter the administrative configuration. Policy settings administratively enforce setting, which restricts user choice.

Preferences can be targeted to specific groups, operating systems, IP addresses, MAC addresses, and more.

Overview of GPO scope and inheritance

GPOs are processed on a client computer in the following order:

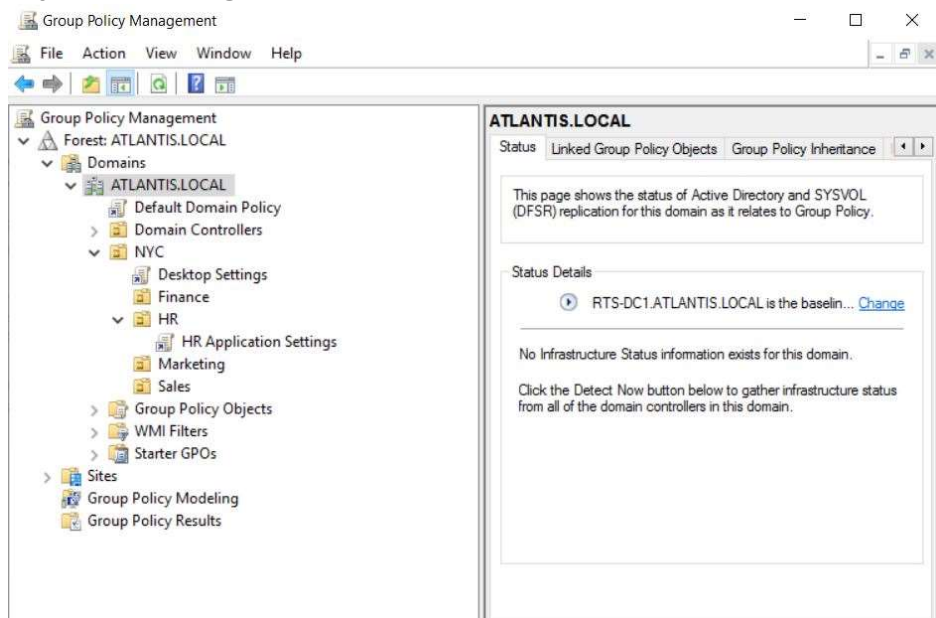
1. Local GPOs
2. Site-level GPOs
3. Domain-level GPOs
4. Organizational Unit GPOs

Altering Group Policy Processing

Block inheritance

Enforced

Security filtering



What is an Active Directory Site?



GPOupdate

What is GPOupdate?

Gpupdate is a command-line utility from Microsoft that comes with all versions of the Windows operating system. It's a utility that controls the application of *group policy objects (GPOs)* on assigned Active Directory computers.

Gpupdate /Force will process all GPOs regardless if they have changed or not

A screenshot of a Windows PowerShell window titled "Administrator: Windows PowerShell". The command prompt shows the following text:

```
PS C:\>  
PS C:\> gpupdate  
Updating policy...  
  
Computer Policy update has completed successfully.  
User Policy update has completed successfully.  
PS C:\>
```

GPOResult

Verify the GPOs that are currently applying to a user and computer account

Examine the settings that apply to the user and computer and determine which GPO applied the setting

Are you ready? CHALLENGE

Create a new virtual machine with the following specifications

VM Name: RTS-SVR1

Memory: 2048MB (enable dynamic memory)

Windows Server 2019 Datacenter (Desktop Experience)

Computer Name: RTS-SVR1

IP Configuration:

IP: 192.168.1.251

Subnet Mask: 255.255.255.0

Gateway: BLANK

DNS: 192.168.1.250

Join RTS-SVR1 to RTSnetworking.com domain

Try it out: Add a shortcut to Internet Explorer favorites

Create a new group policy object (GPO) named "**Sales Website**", link the GPO to **Sales** OU

Define a **GPO preference** under **User Configuration**

Add the URL <https://rtsnetworking.com> Named "**RTS**" as a shortcut

Use Item-Level targeting to target the **Sales-Users** group

Verify that **Bob Ross** is a member of the **Sales-Users** group

*Make **Bob Ross** a member of the **Domain Admins** group (for future exercises)

Sign-in to **RTS-SVR1** as **Bob Ross**

Open **Internet Explorer** and verify that **RTS** is listed in favorites



Module 3: Managing DHCP and DNS

Module overview

Lessons:

- Deploying and managing DHCP
- Deploying and managing DNS

Lesson 1 overview

Topics:

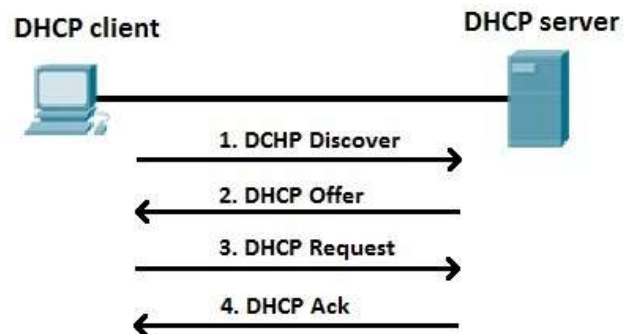
- Overview of the DHCP role
- Install and configure the DHCP role
- Configure DHCP scopes
- DHCP AD DS authorization
- DHCP Failover for high availability

Overview of the DHCP role

DHCP automates management of IP configuration on clients and devices

DHCP lease renewal is attempted at:

- Startup
- 50% of lease time
- 87.5% of lease time



Install and configure the DHCP role

To install the DHCP role:

- **Windows Admin Center > Roles and Features**
- **Server Manager**
- **Add-WindowsFeature DHCP -IncludeManagementTools**

To manage a DHCP server by using Windows Admin Center, you must install the DHCP PowerShell tools

Install and configure the DHCP role

DHCP local security groups:

- DHCP Administrators
- DHCP Users

To create the DHCP local security groups:

- **Server Manager > Post-Install Configuration Wizard**
- **Add-DhcpServerSecurityGroup -Computer DhcpServerName**

Configure DHCP scopes

Properties of a DHCP scope:

- Name (mandatory)
- IP address range (mandatory)
- Subnet mask (mandatory)
- Exclusions
- Delay
- Lease duration
- Options
- Activation

Option code	Name
3	Router (Default Gateway)
6	DNS servers
15	DNS domain name

DHCP AD DS authorization

A DHCP server on Windows Server must be authorized in AD DS to lease IP addresses:

- To authorize a DHCP server by using Windows PowerShell, run:
Add-DHCPServerinDC <name or IP address of DHCP server>

A standalone server with DHCP will not lease IP addresses if an authorized DHCP server is detected

Non-Windows DHCP servers and devices will function regardless of authorization

DHCP High availability

Split scopes:

- Involve two DHCP servers that are configured with non-overlapping scopes

DHCP failover:

- Scopes are replicated from one DHCP to another DHCP partner
- Strongly preferred to implement high availability for DHCP

Failover configuration modes:

- Load balance
- Hot standby

Lesson 2 overview

Topics:

- DNS components
- DNS records
- DNS zones
- Create records in DNS
- Configure DNS zones
- DNS forwarding
- Integrating DNS with Active Directory

DNS components

DNS domain names:

- Are a portion of DNS namespace
- Can be public or private

DNS servers:

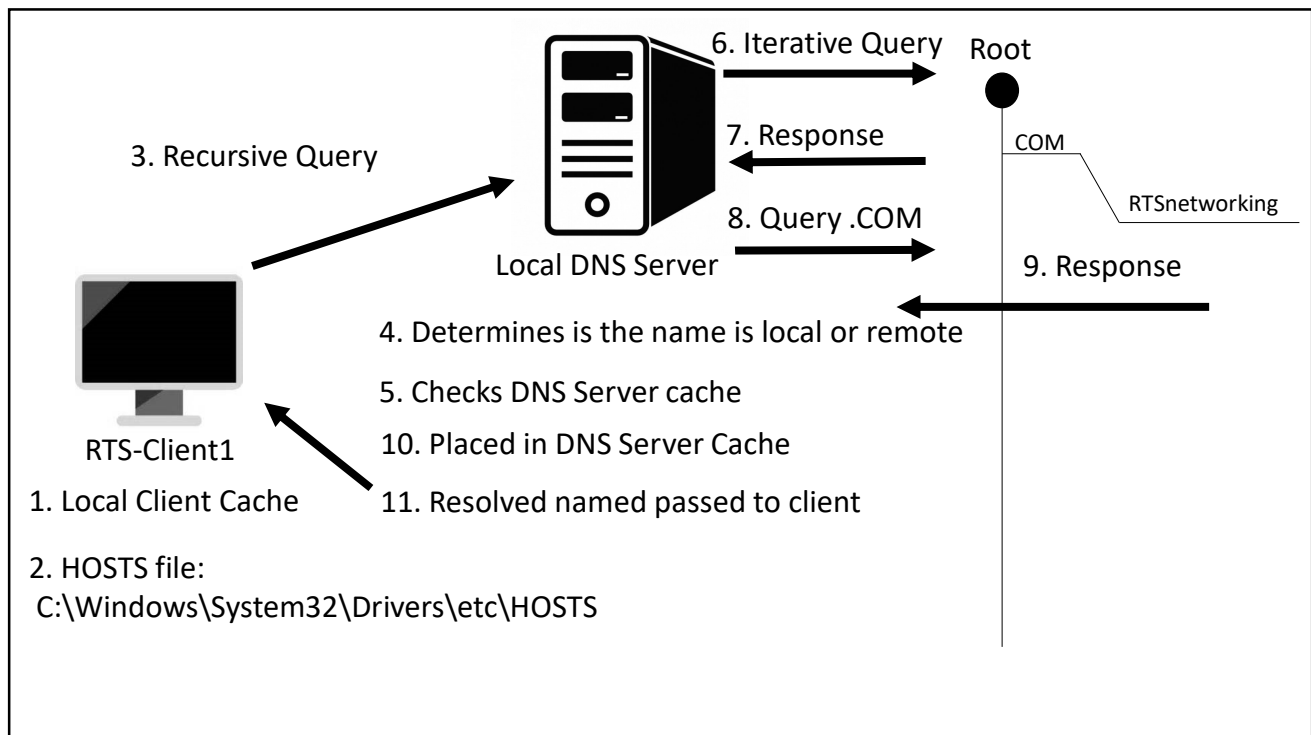
- Respond to name resolution requests
- Stores resource records locally in a database on the DNS server

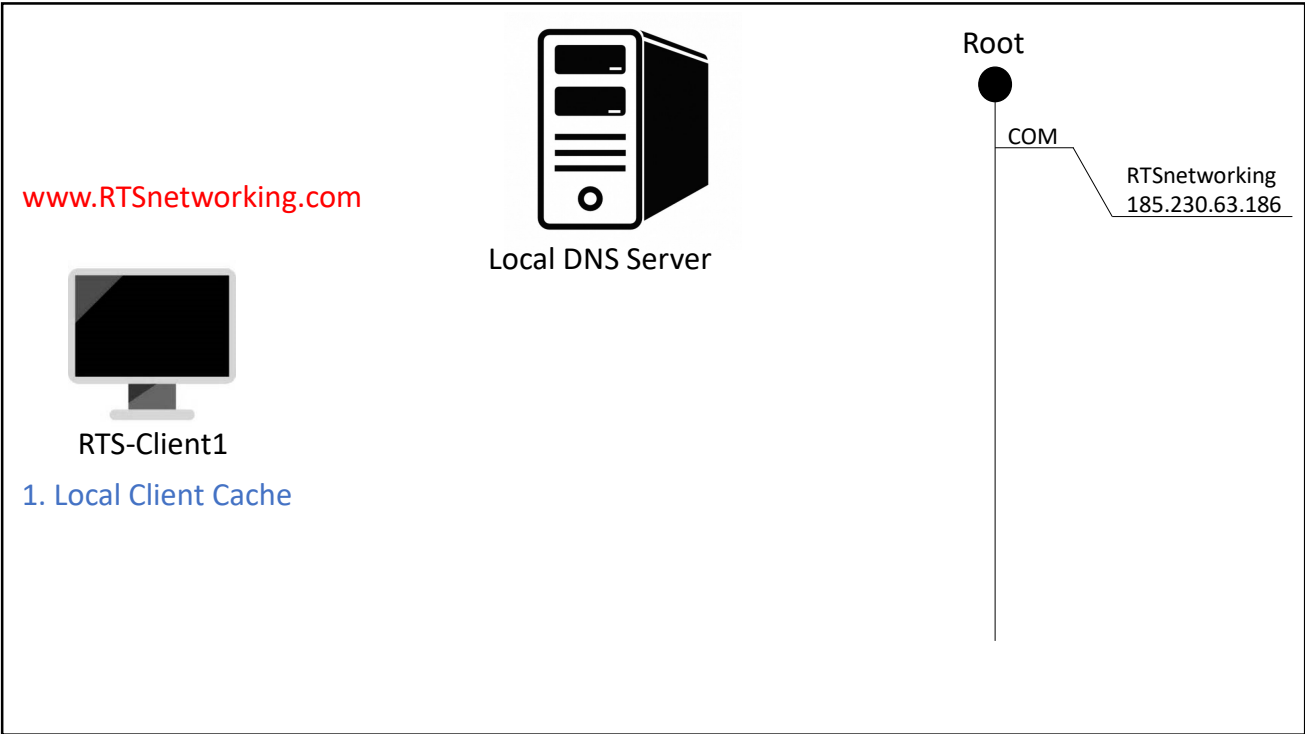
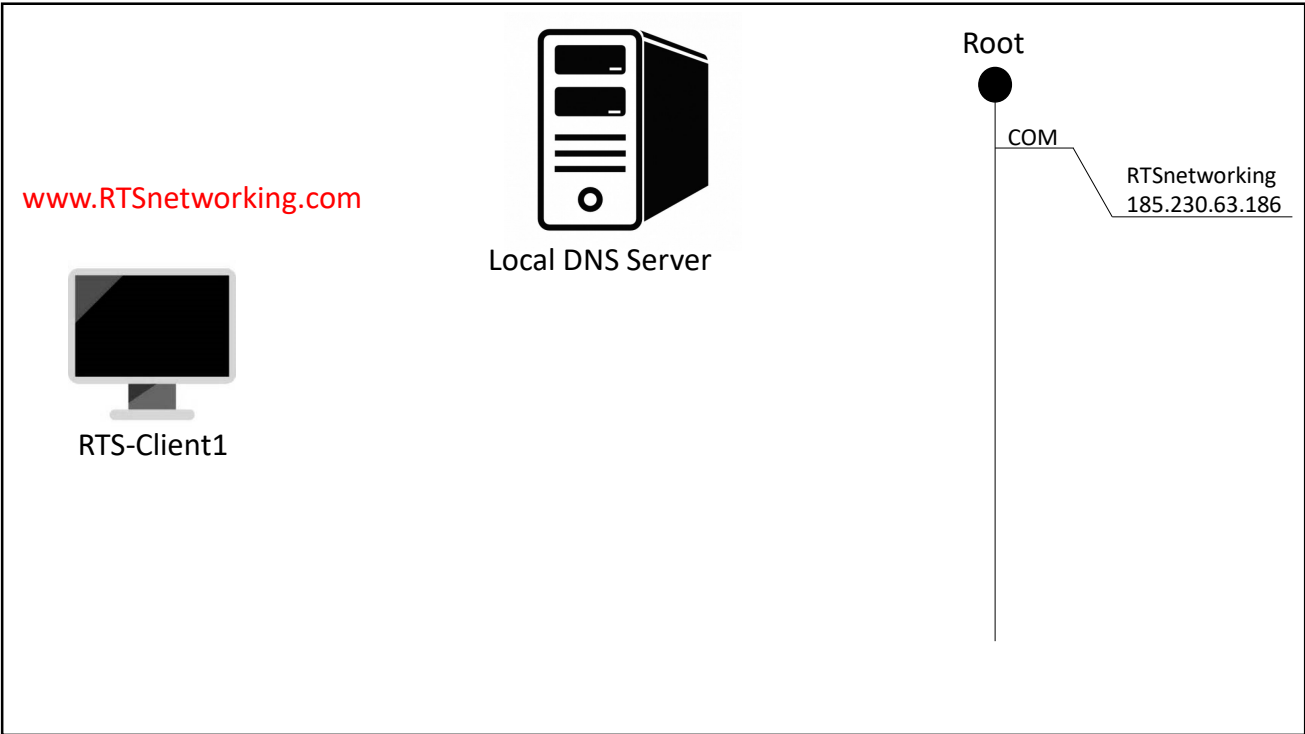
DNS zones and resource records:

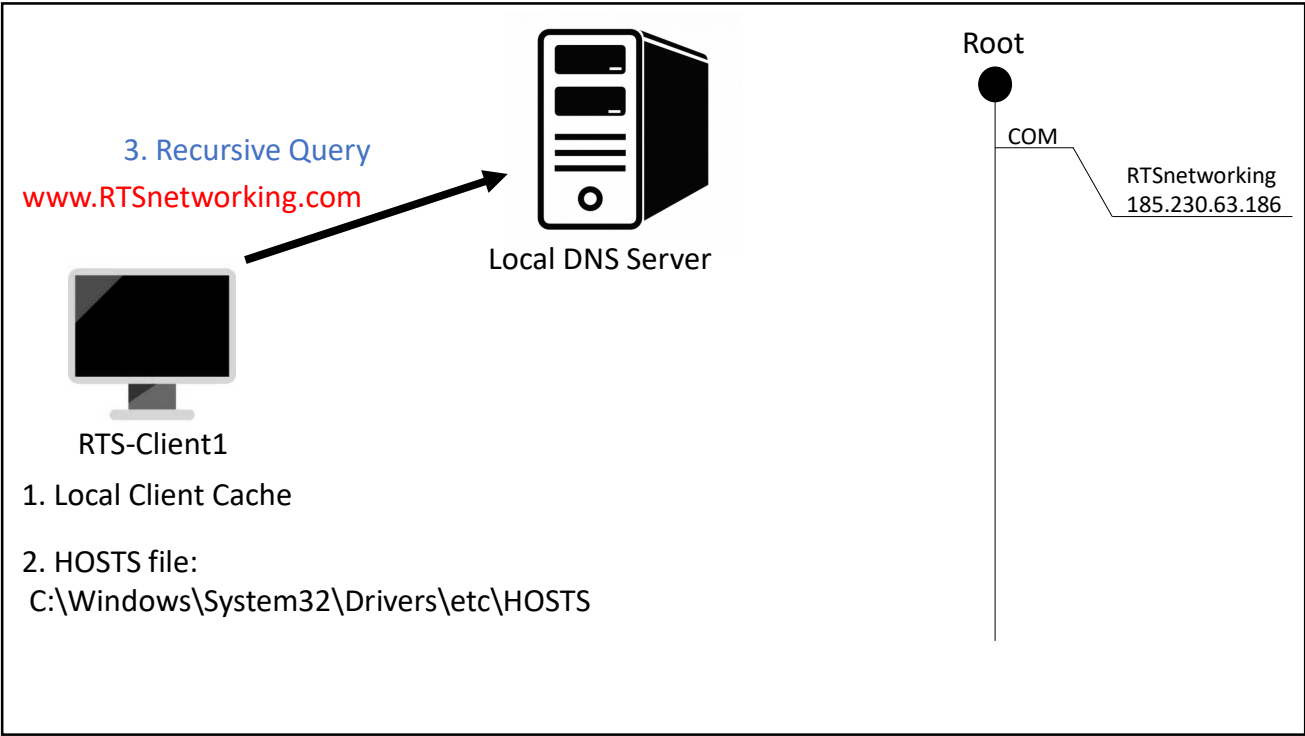
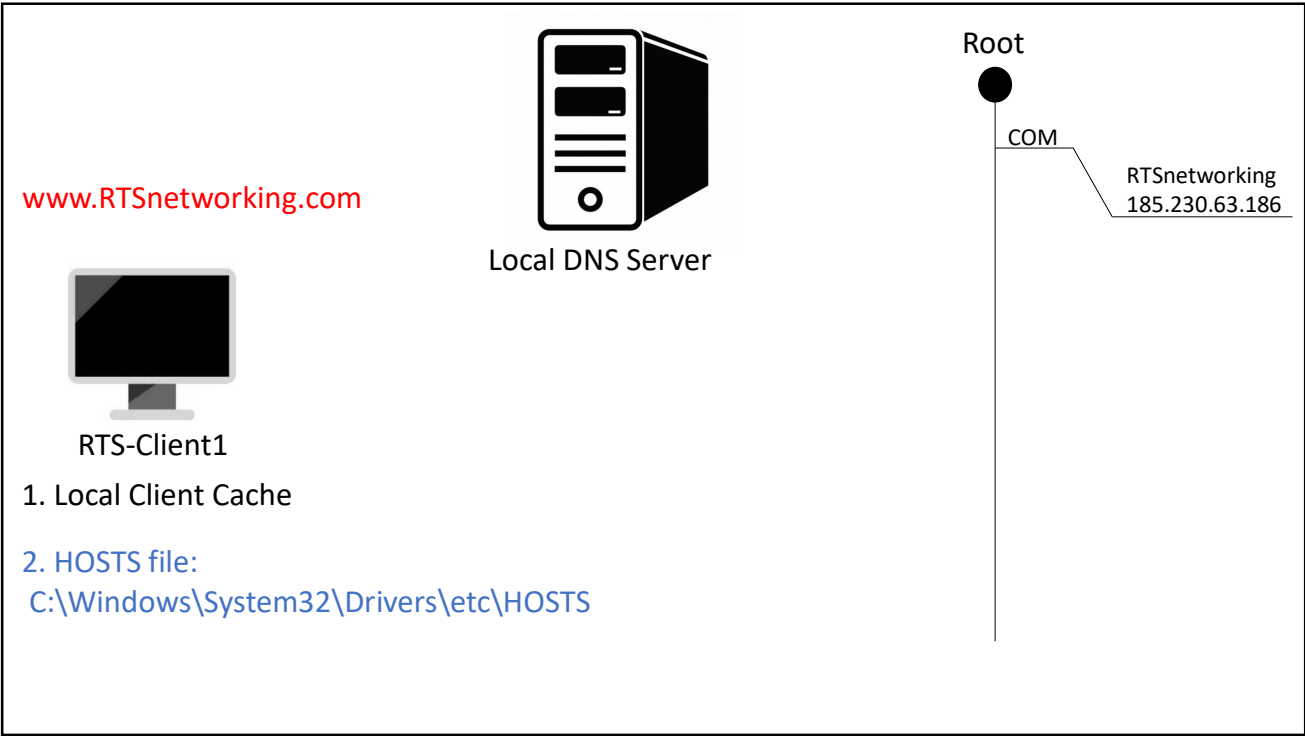
- A zone is a local copy of a DNS namespace on a DNS server
- Resource records are created and stored in a zone

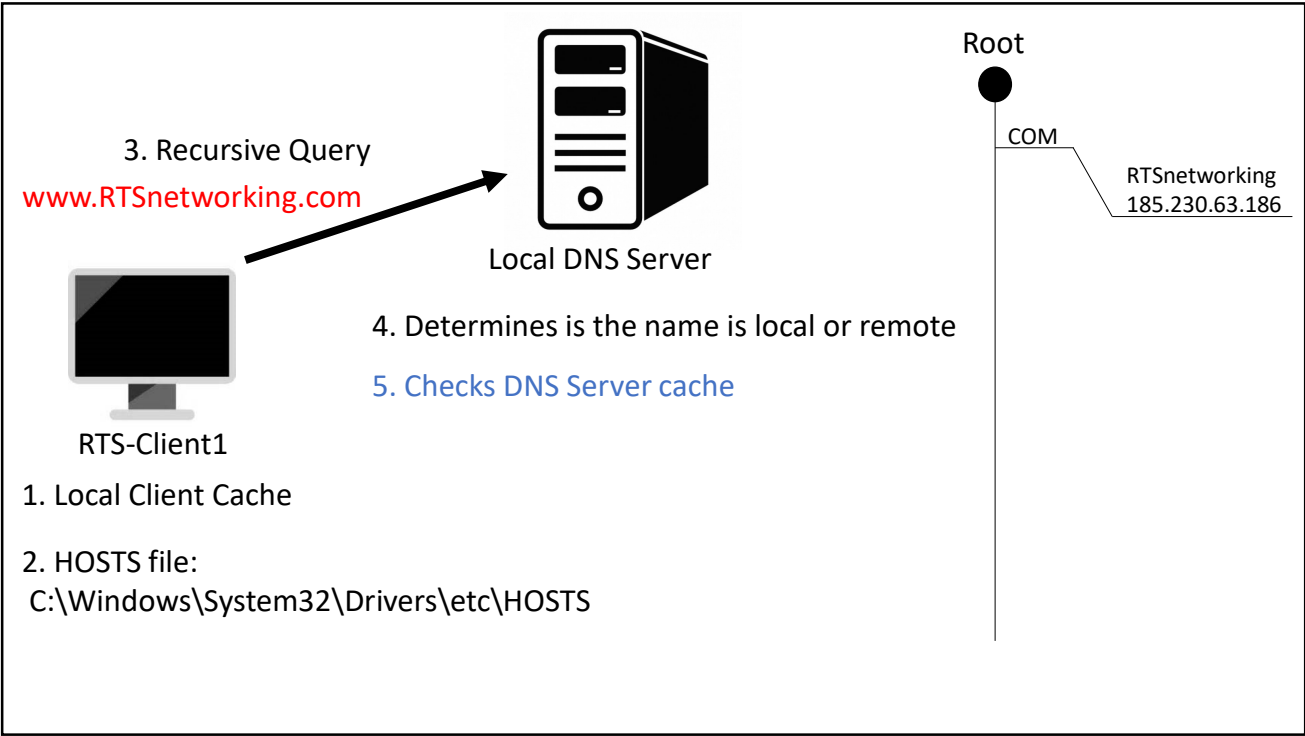
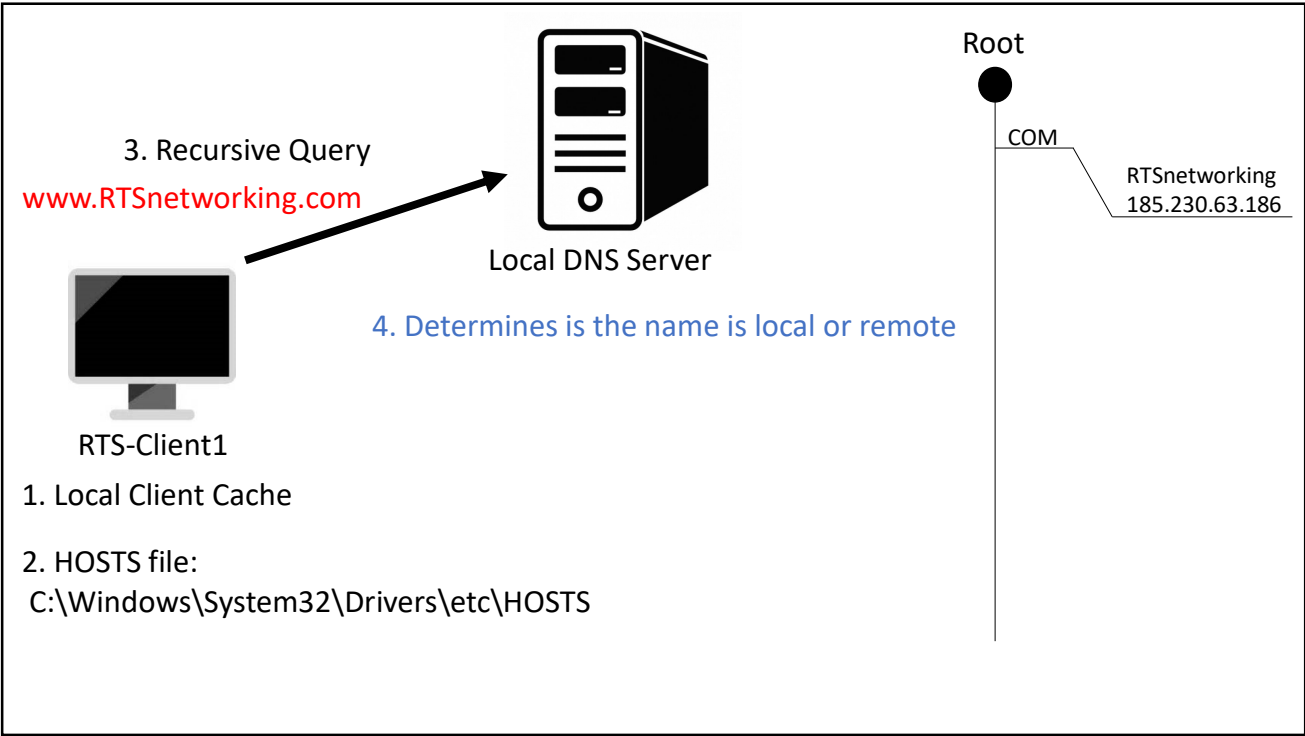
DNS resolvers:

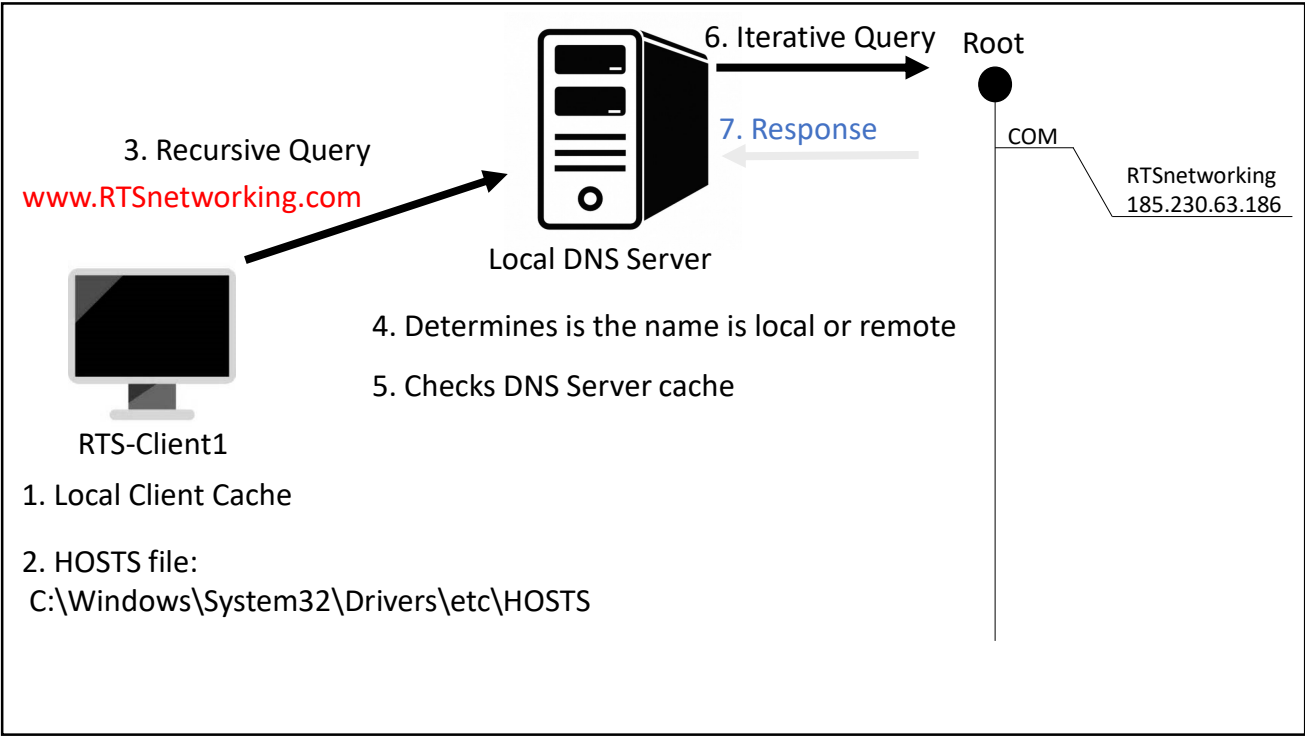
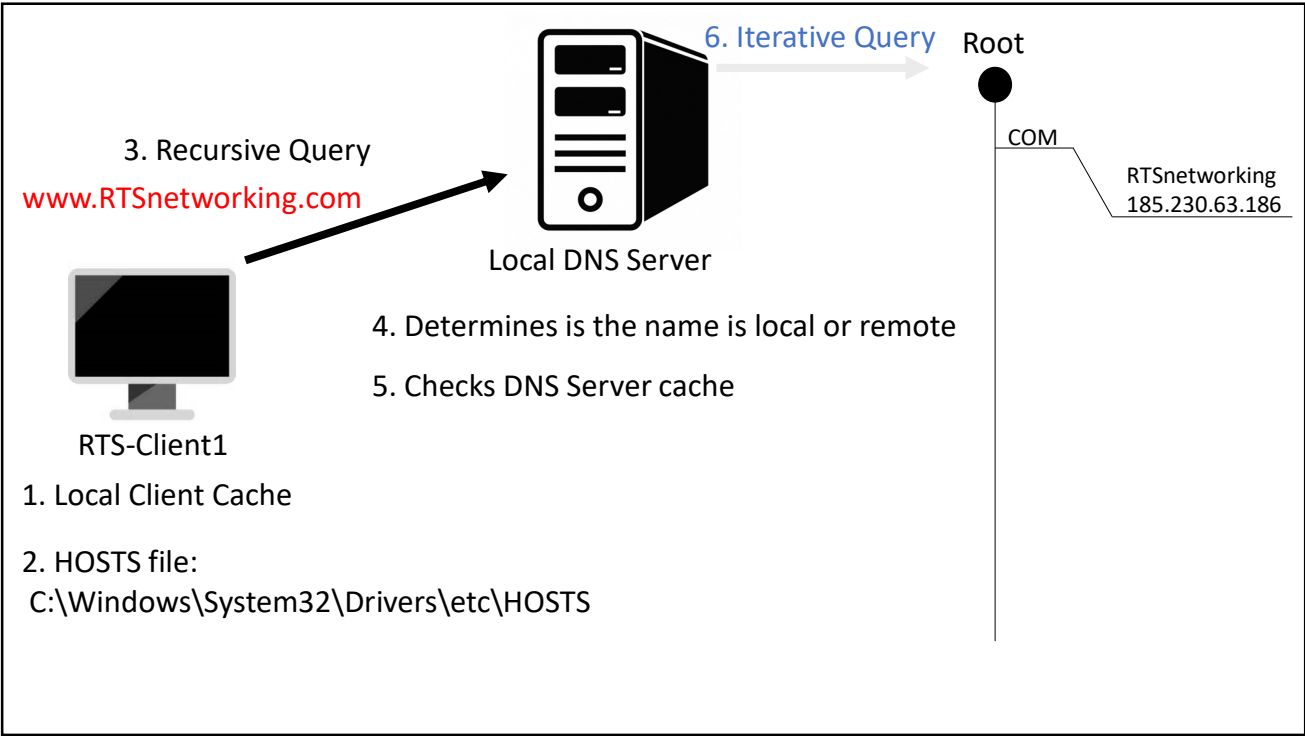
- Request DNS information from DNS servers
- Cache results

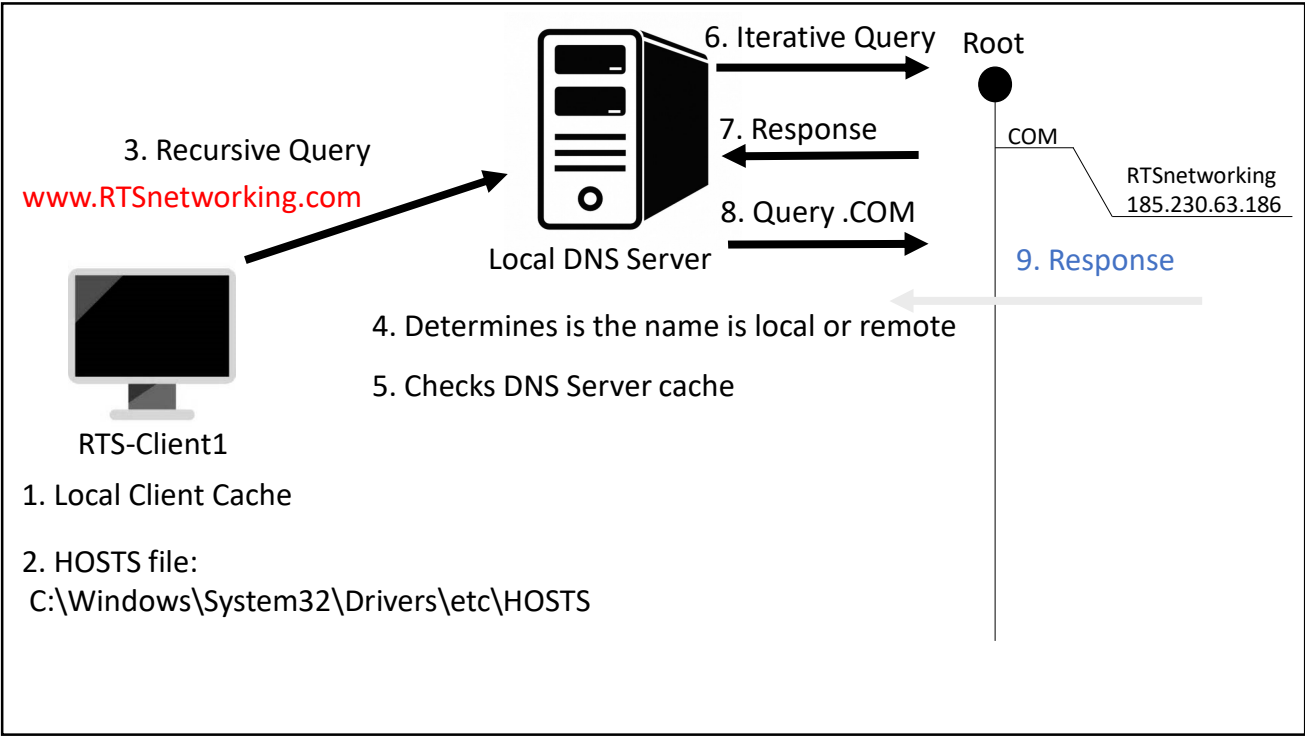
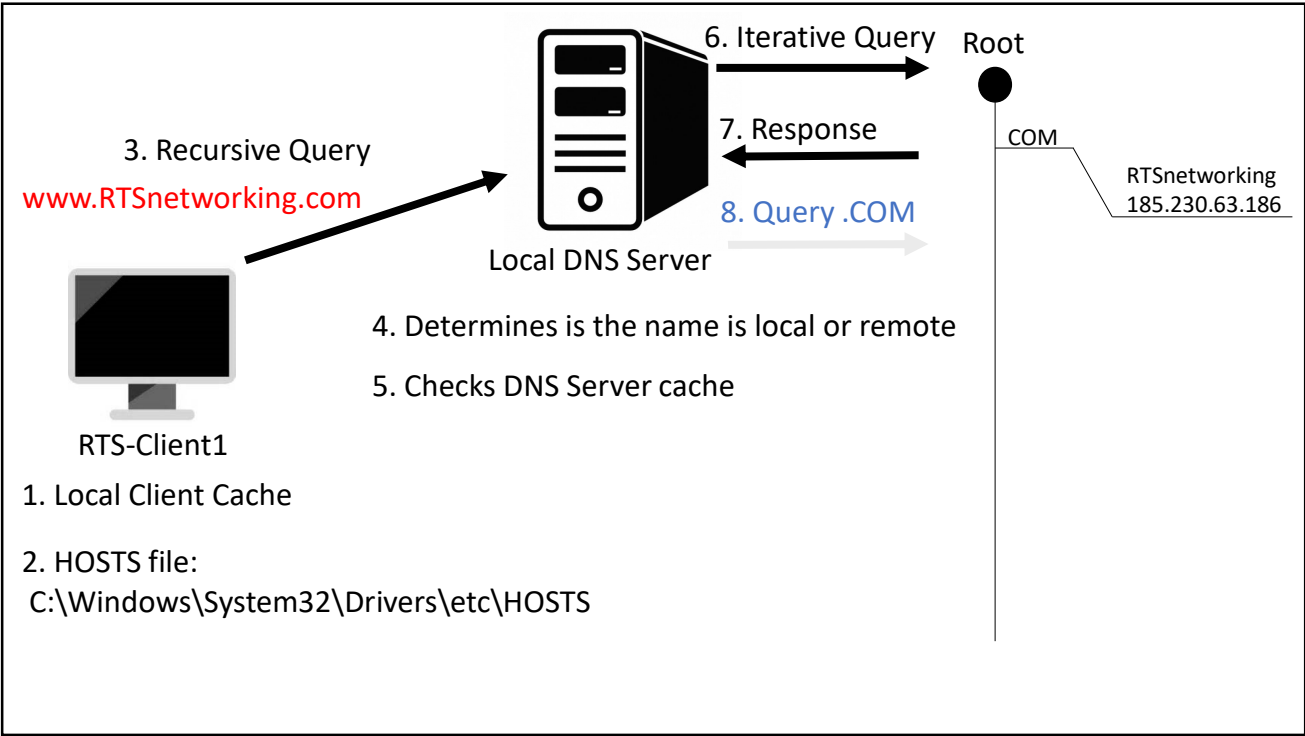


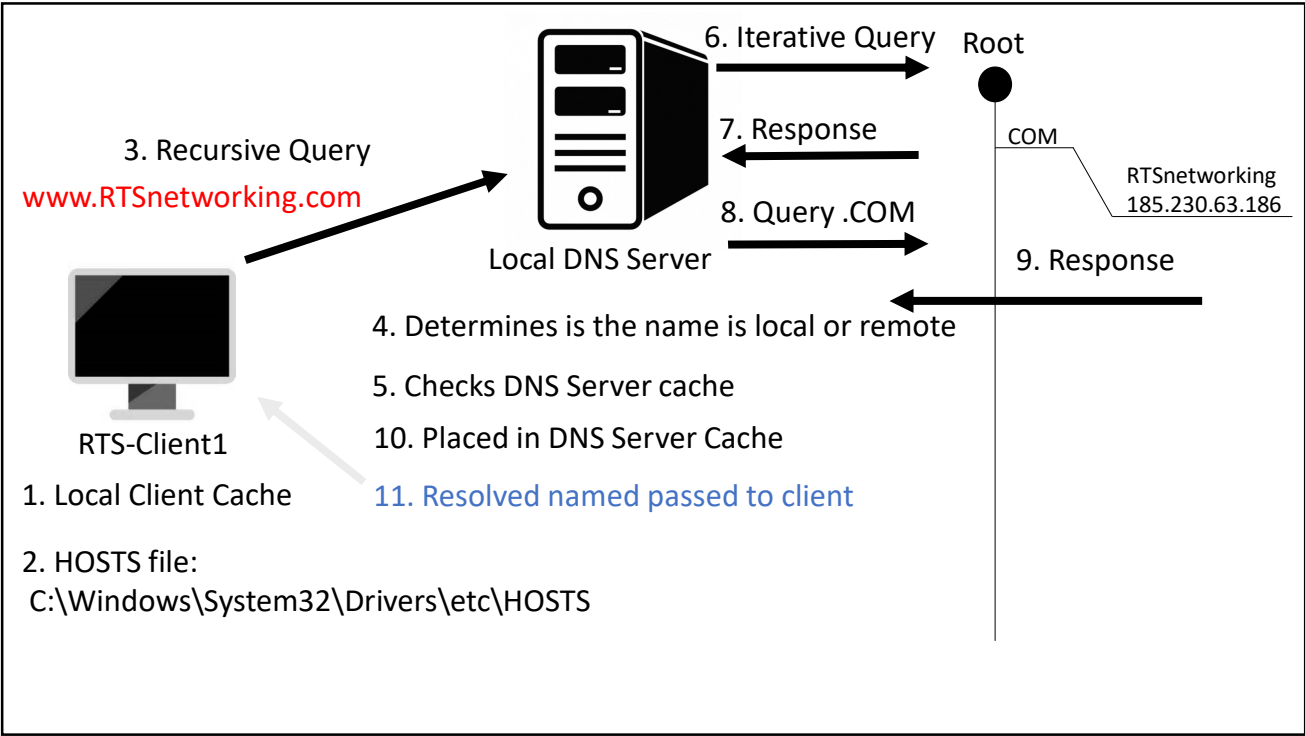
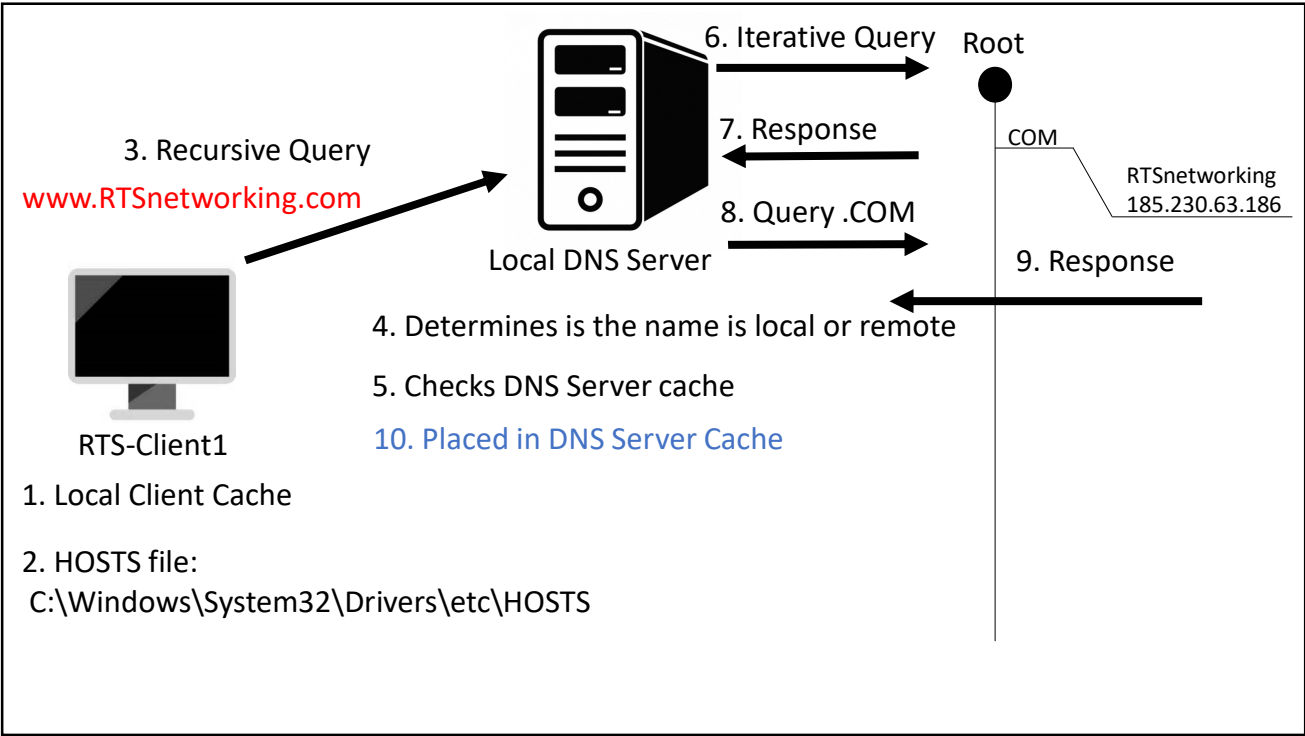












DNS records

Forward lookup zones include:

- Host (A)
- Host (AAAA)
- Alias (CNAME)
- Service location (SRV)
- Pointer (PTR)

Create records in DNS

Manual creation methods:

- Windows Admin Center
- DNS manager
- Windows PowerShell

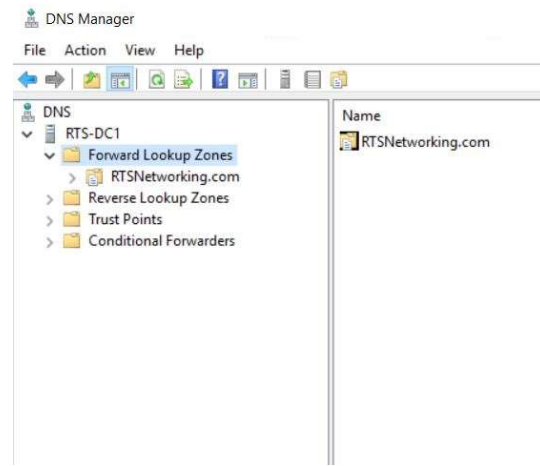
Dynamic creation:

- Clients register name and IP address in a zone

DNS zones

A DNS zone is the portion of a DNS namespace hosted on a DNS server:

- Forward lookup zones:
 - Resolve names to IP addresses
 - Can contain many other record types
- Reverse lookup zones:
 - Resolve IP addresses to names
 - Are in the **in-addr.arpa** namespace



DNS zones

Primary zones:

- Are authoritative for a portion of a DNS namespace
- Are where resource records are created

Secondary zones

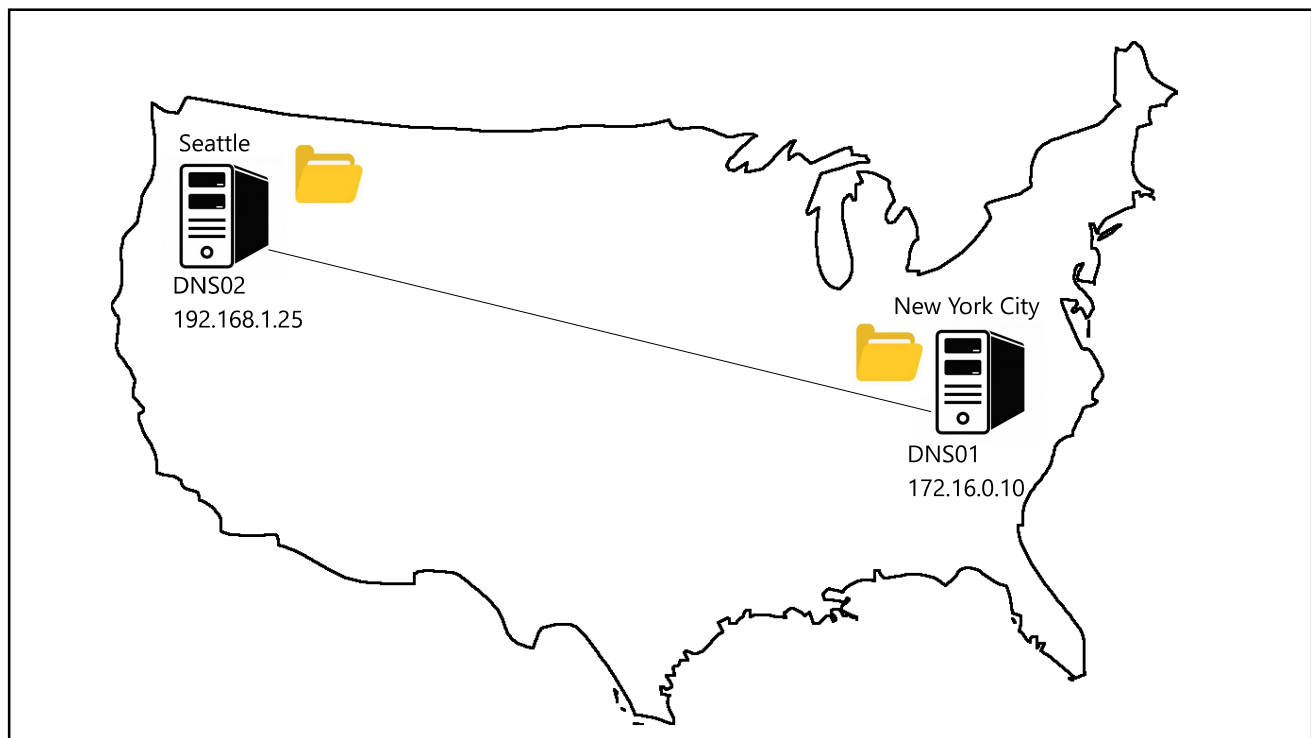
- Read-only copies of primary zones

Stub zones

- contain only the records required to locate and communicate with name servers

Active Directory-integrated zones

- Can only reside on domain controllers
- Replicates with active directory



DNS Forwarding

Forwarders:

- Receive DNS requests, and forward requests for zones for which it is not authoritative
- Are common for external name resolution

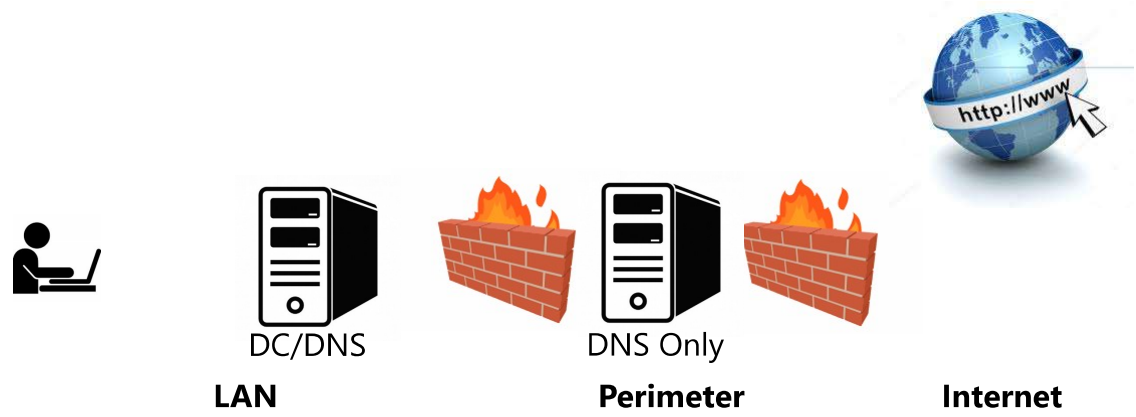
Conditional forwarders:

- Forward requests for a specific domain name
- Typical between partners and trusted organizations

Stub zones:

- Have a similar role to conditional forwarders
- Are used within the same company
- Requires configuration on both DNS Servers

DNS Forwarding





Module 4: Managing File Servers and Storage

Module Overview

This module describes how to configure file servers and storage in Windows Server:

- Lessons:
 - Volumes and file systems in Windows Server
 - Implementing sharing in Windows Server
 - Implementing Storage Spaces in Windows Server
 - Implementing Data Deduplication
 - Deploying Distributed File System

Lesson 1 Overview

This lesson describes file systems and volumes in Windows Server:

- Topics:
 - Overview of file systems in Windows Server
 - Overview of disk volumes

Overview of file systems in Windows Server

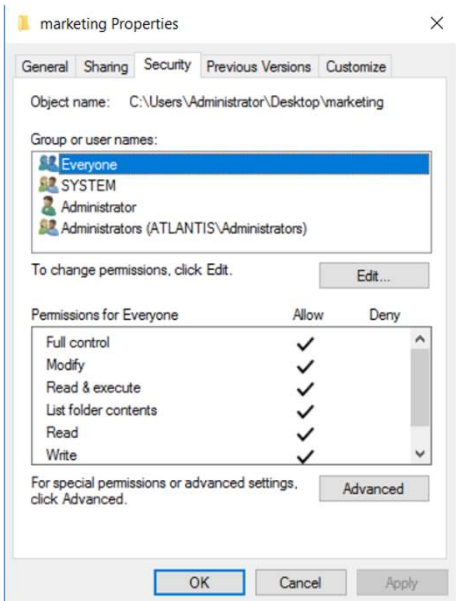
When selecting a file system, consider the differences between FAT, NTFS file system, and ReFS:

- FAT/FAT32 provides:
 - Basic file system
 - No Security
 - exFAT developed for flash drives
- NTFS provides:
 - Auditing
 - Security (permissions and encryption)
 - Compression
- ReFS provides:
 - Backward compatibility support for NTFS
 - Enhanced data verification and error correction
 - Support for larger files, directories, and volumes



File and Folder Level Security

Read
Read and Execute
Write
Modify
Full Control
List Folder Contents



Managing NTFS Permissions



Sales-Reports

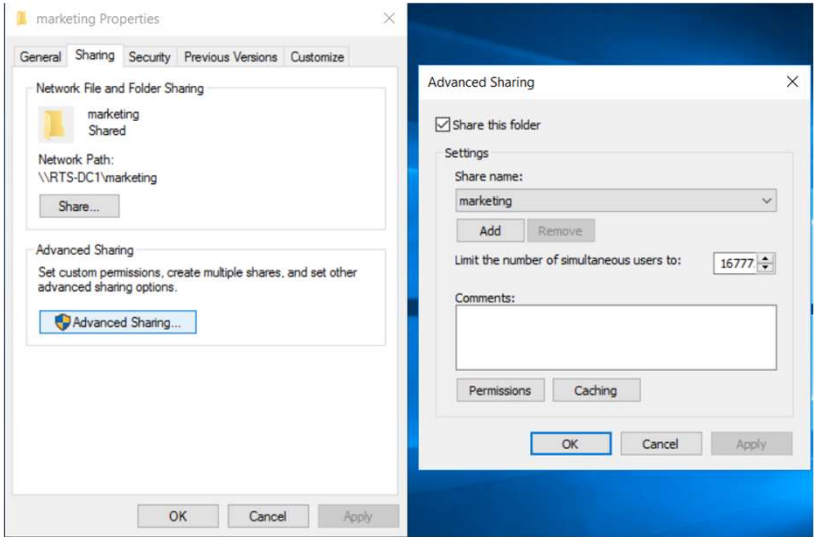


Permissions	Allow	Deny
Full Control		
Modify	☒	
Read and execute		
List Folder Contents		
Read	☒	
Write		

Managing Shared Folders

Share Permissions:

- Read
- Change
- Full Control



Inheritance

Permissions

Full Control
Modify
Read and execute
List Folder Contents
Read
Write

Sales-Reports



Jan-Reports

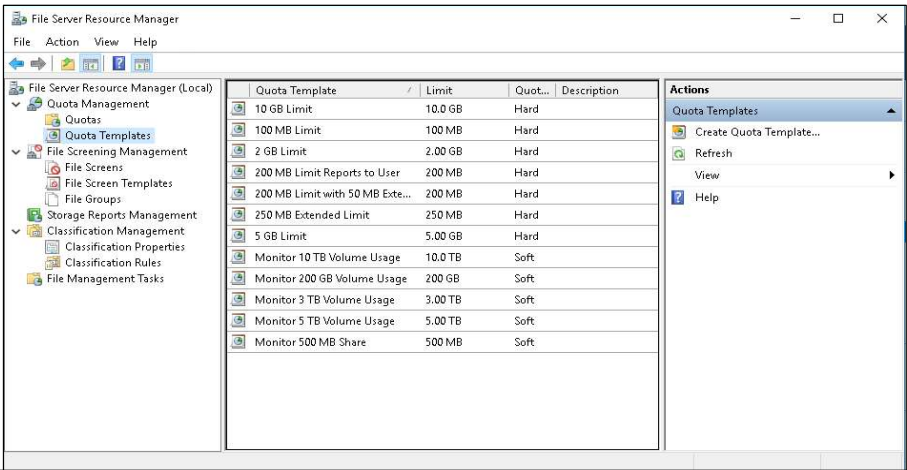


Permissions

Full Control
Modify
Read and execute
List Folder Contents
Read
Write

Overview of File Server Resource Manager

- Create quotas to monitor and limit the amount of space consumed
- Use a file screen to monitor and block files based on the file extension



Quotas and File Screens

User-Data



Quota:
Limit data to 10GB

File Screen:
Block music and video files
from being saved

Overview of disk volumes

When selecting a type of disk for use in Windows Server, you can choose between:

- Basic disk
- Dynamic disk



Overview of disk volumes

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- Dynamic disk



Overview of disk volumes

When selecting a type of disk for use in Windows Server, you can choose between:

- Basic disk
- Dynamic disk

In Windows Server, if you are using dynamic disks, you can create a number of different types of disk volumes:

- Simple volumes
- Spanned volumes
- Striped volumes
- Mirrored volumes
- RAID-5 volumes



Overview of disk volumes

When selecting a type of disk for use in Windows Server, you can choose between:

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- **Mirrored volumes**
- RAID-5 volumes



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In Windows Server, if you are using dynamic disks, you can create a number of different types of disk volumes:

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- Striped volumes
- Mirrored volumes
- **RAID-5 volumes**



Lesson 2: Overview

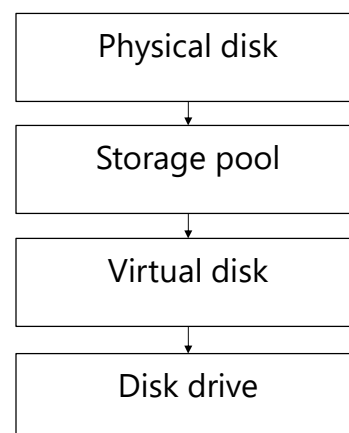
This lesson describes how to deploy Storage Spaces technology in Windows Server

- Topics:
 - What are Storage Spaces?
 - Components and features of Storage Spaces
 - Storage Spaces usage scenarios
 - Provisioning a Storage Space

What are Storage Spaces?

Use Storage Spaces to:

- Add physical disks of any type and size to a storage pool
- Create highly-available virtual disks from the pool:
 - To create a virtual disk, you need:
 - One or more physical disks
 - A storage pool that includes the disks
 - Virtual disks (or storage spaces) that are created with disks from the storage pool
 - Disk drives that are based on virtual drives



Storage Spaces usage scenarios

Storage Spaces was designed to enable storage administrators to:

- Implement and easily manage scalable, reliable, and inexpensive storage
- Use inexpensive storage with or without external storage
- Combine multiple drives into storage pools that administrators can manage as a single entity
- Implement different types of storage in the same pool
- Grow storage pools as required
- Provision storage as required from existing storage pools
- Designate specific drives as hot spares

Demo

Storage space

Lesson 3: Overview

This lesson describes how to implement the Data Deduplication feature:

- Topics:
 - Data Deduplication components
 - Data Deduplication process
 - Deploying Data Deduplication
 - Backup and restore considerations with Data Deduplication

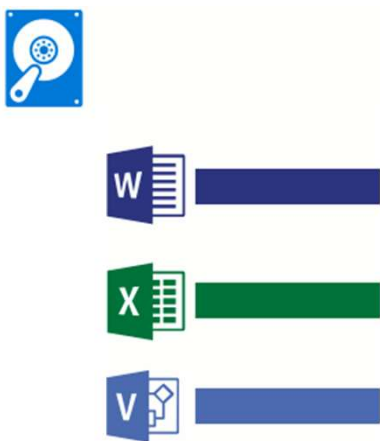
Data Deduplication

- **Data Deduplication**, often called **Dedup** for short, is a feature that can help reduce the impact of redundant data on storage costs.
- When enabled, Data Deduplication optimizes free space on a volume by examining the data on the volume by looking for duplicated portions on the volume.
- Duplicated portions of the volume's dataset are stored once and are compressed for additional savings.

Scenario	Content	Typical space savings
User documents	Office documents, photos, music, videos, etc.	30-50%
Deployment shares	Software binaries, cab files, symbols, etc.	70-80%
Virtualization libraries	ISOs, virtual hard disk files, etc.	80-95%
General file share	All the above	50-60%

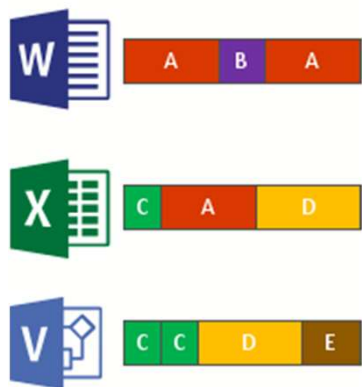
Data Deduplication

- 1. Scan the file system for files meeting the optimization policy



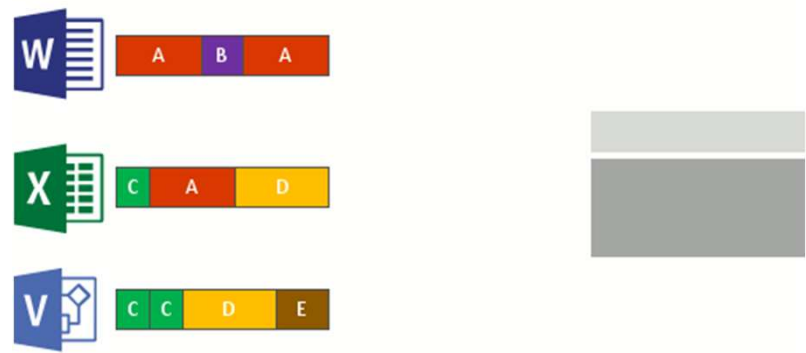
Data Deduplication

- 2. Break files into chunks



Data Deduplication

- 3. Identify unique chunks
- 4. Place chunks in the chunk store and compress



Data Deduplication

- 5. Replace the original file stream of now optimized files with a reparse point to the chunk store



Deploy Data Deduplication

Prior to installing and configuring Data Deduplication in your environment, plan your deployment using the following steps:

1. Determine target deployments (the drive to which you want to deploy dedup)
2. Determine which volumes are candidates for deduplication
3. Evaluate savings with the **Deduplication Evaluation Tool**
4. Plan the rollout and deduplication policies

Deploy Data Deduplication

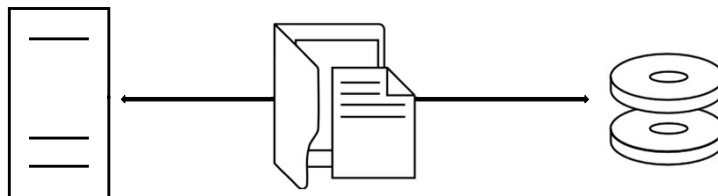
After completing your planning, deploy Data Deduplication to a server in your environment by performing the following steps:

1. Install Data Deduplication components on the server
2. Enable Data Deduplication
3. Configure Data Deduplication jobs
4. Configure Data Deduplication schedules

Demo

deduplication

Backup and restore considerations with Data Deduplication



One of the benefits of using Data Deduplication is that backup and restore operations typically are faster

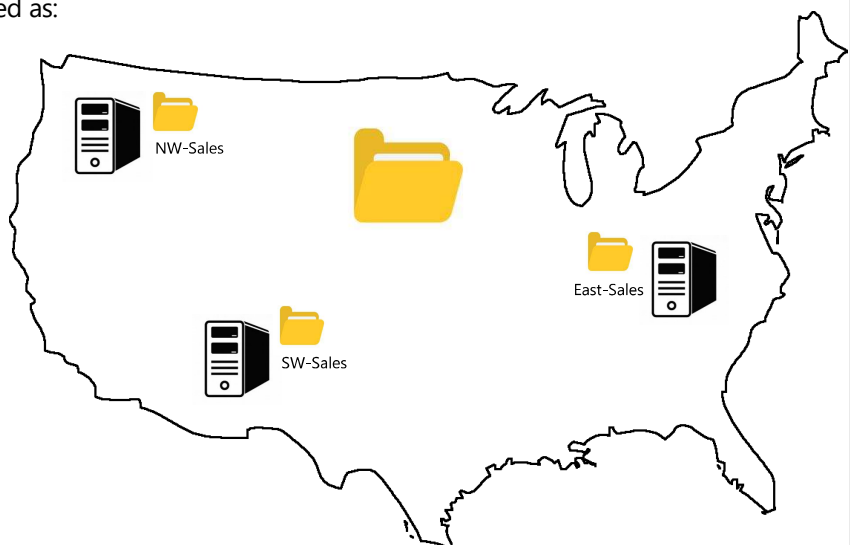
Lesson 4: Overview

This lesson describes how to manage DFS databases

- Topics:
 - DFS namespace
 - DFS Replication
 - Implement DFS namespace and replication solutions

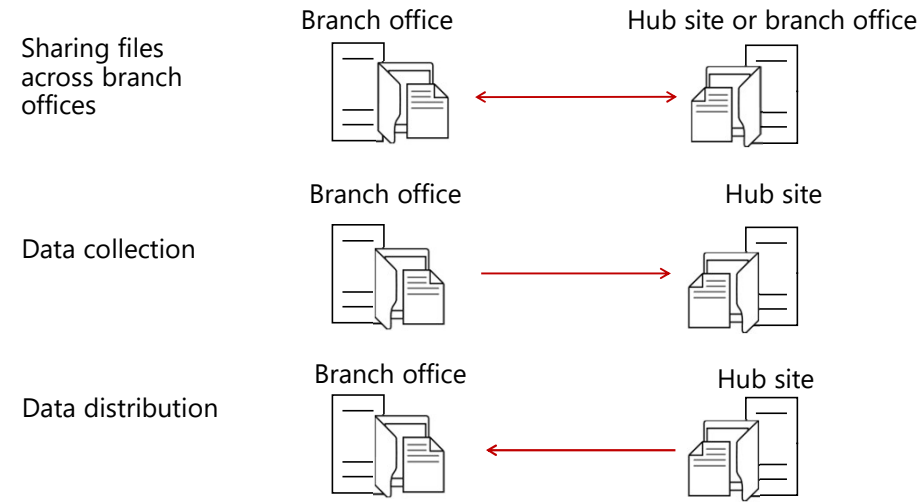
DFS namespaces

- DFS namespaces can be configured as:
 - Domain-based namespaces
 - Standalone namespaces



DFS Replication

Three DFS scenarios:



Demo

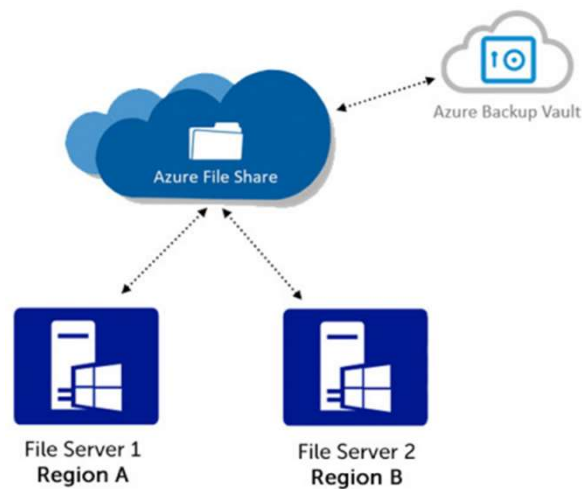
DFS N and DFS R

Overview of Azure File Sync

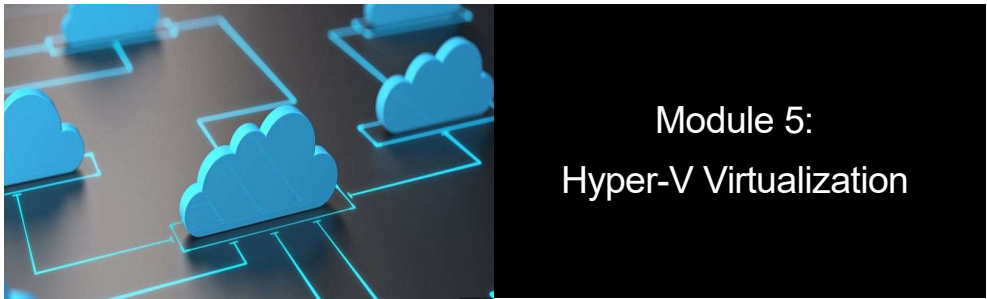
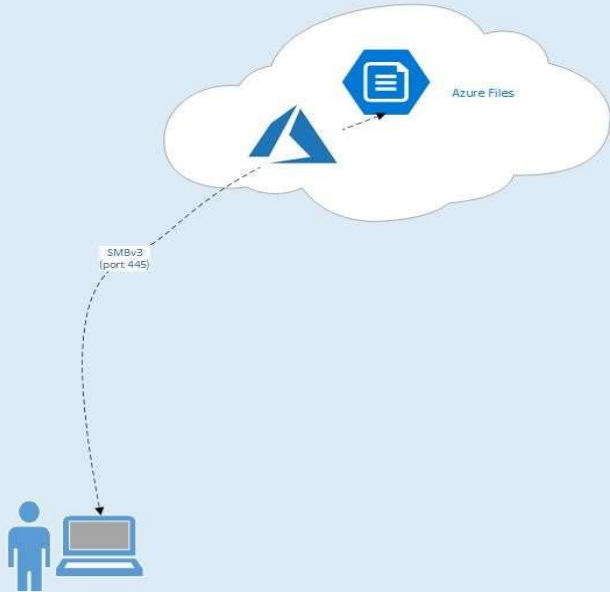
Azure File Sync:

- Automatically replicates local file shares to a central Azure storage location
- Administrators continue to support local data access
- Automatically backs up disparate branch office data
- Allows cloud tiering—non-used data moves off premises to Azure
- Uses SSL encryption through well-known port number 443

Azure File Sync



Azure File Sync



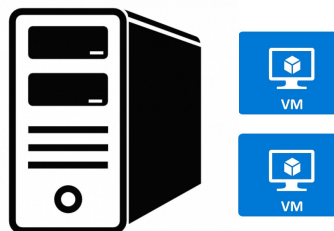
Lesson 1 Overview

This lesson provides an overview of Hyper-V :

- Topics:
 - Understanding Hyper-V
 - Hyper-V manager
 - Best practices for Hyper-V configurations

Overview of Hyper-V

- Hyper-V is a hardware virtualization server role available for Windows Server
- Provides a software layer known as the *Hypervisor*, used to control access to physical hardware
- Supports many types of guest operating systems including:
 - All supported Windows versions
 - Linux
- General Hyper-V features can be grouped as follows:
 - Management and connectivity
 - Portability
 - Disaster recovery and backup
 - Security
 - Optimization



Overview of Hyper-V

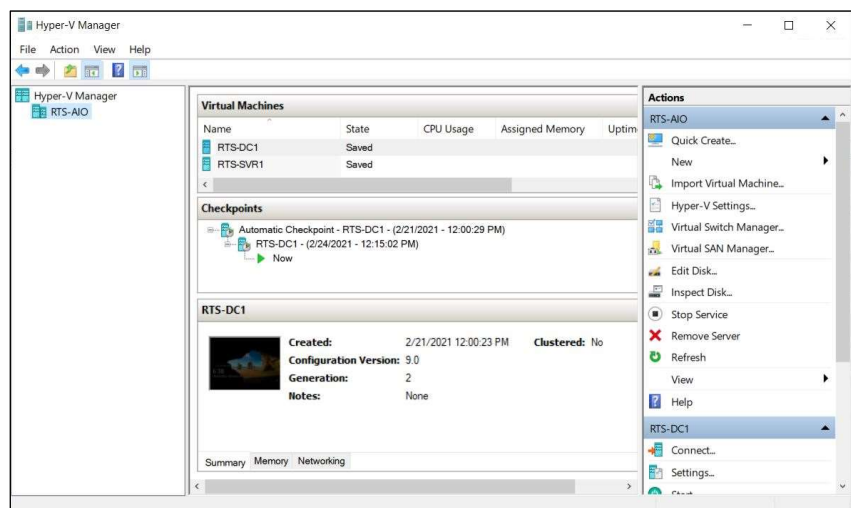
- System requirements for installing the Hyper-V server role include:
 - A 64-bit processor
 - Sufficient memory
 - Intel Virtualization Technology (Intel VT) or Advanced Micro Dynamics (AMD) Virtualization (AMD-V) enabled

To verify you meet the requirements, run **MSINFO32**

- Methods to install the Hyper-V server role include:
 - Server Manager
 - **Install-WindowsFeature** PowerShell cmdlet

Overview of Hyper-V Manager

- A graphical user interface used to manage both local and remote Hyper-V host machines
- Other management tools include:
 - Windows PowerShell
 - Windows Admin Center



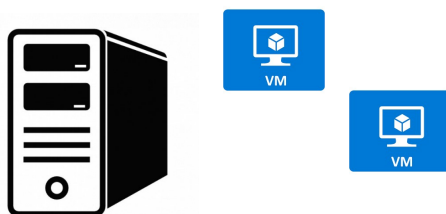
Best practices for configuring Hyper-V hosts

- Consider the following when provisioning Windows Server as a Hyper-V host:
 - Provision the host with adequate hardware
 - Deploy virtual machines on separate disks, solid state drives
 - Do not collocate other server roles
 - Manage Hyper-V remotely
 - Run Hyper-V by using a Server Core configuration

Overview of nested virtualization

- Provides the ability to install the Hyper-V role within a guest virtual machine
- Requirements:
 - Both the Hyper-V host and the guest virtual machine must be Windows Server 2016 or later
 - Sufficient amount of static RAM

Set-VMProcessor -VMName <VMName> -ExposeVirtualizationExtensions \$true



demo

Nested virtualization

Migration to Azure VMs

- **Azure Migrate** can be used to migrate on-premises workloads, apps, and virtual machines
- **Azure Migrate** provides the following benefits:
 - A single migration platform
 - Assessment and migration tools
 - **Azure Migrate: Server Assessment**
 - **Azure Migrate: Server Migration**
 - Ability to assess and migrate multiple object types:
 - Servers
 - Databases
 - Web applications
 - Virtual desktops
 - Data



demo

Azure VM

Lesson 2 Overview

This lesson describes configuring virtual machines:

- Topics:
 - Configuring VMs
 - VM Settings
 - Virtual Hard Disks
 - Virtual Networking
 - Managing Checkpoints

VM generation versions

- Generation 1 VMs:
 - Support 32 and 64-bit operating systems
 - Only support boot volumes a maximum of 2 TB
 - Supports legacy BIOS
- Generation 2 VMs:
 - Support only 64-bit operating systems
 - Support secure boot and shielded VMs
 - Support boot volumes a maximum of 64 TB
 - Supports Unified Extensible Firmware Interface (UEFI)

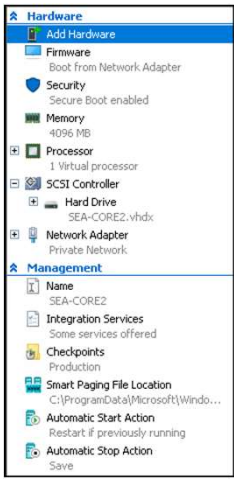
VM settings

- VM settings are grouped into two main areas:
 - Hardware
 - Management
- Available hardware components depend on the generation version of the VM

Generation 1 settings



Generation 2 settings

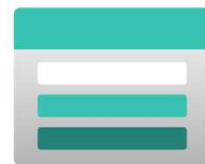


demo

New vm

Storage options in Hyper-V

- Consider the following factors when planning storage for virtual hard disks:
 - High-performance connection to storage
 - Redundant storage
 - High-performance storage
 - Adequate growth space



Virtual hard disk formats and types

- Virtual hard disk formats include:
 - VHD
 - Up to 2040 GB in size
 - Typically used to support older Hyper-V versions
 - VHDX:
 - Up to 64 TB in size
 - Recovery from corruption issues
 - Supports larger block size resulting in increased performance
- Use the **Edit Virtual Hard Disk Wizard** to convert between hard disk formats
- Various tools can be used to create and manage virtual hard disks:
 - Hyper-V Manager
 - Disk Management/Diskpart
 - PowerShell (New-VHD)
 - Windows Admin Center

Virtual hard disk formats and types

Type of disc	Description
Fixed	Allocates all of the hard disk space immediately
Dynamic	The disk only uses the amount of space that needs to be allocated, and it grows as necessary
Differencing	Associated with another virtual hard disk in a parent-child configuration. Any changes made to the differencing disk does not affect the parent disk.
Pass through	Allows the virtual machine to connect directly to an Internet Small Computer Systems Interface (iSCSI) (logical unit number) LUN or a physical disk attached on the host machine



Demo disk management

Overview of Hyper-V networking

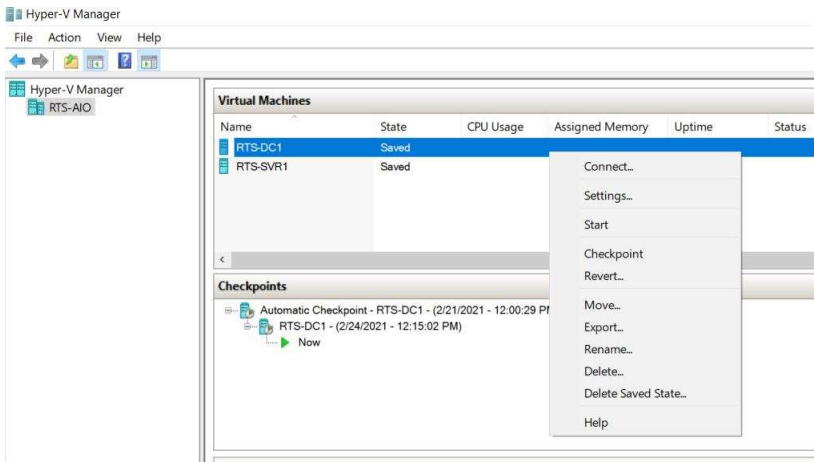
- Hyper-V supports three types of virtual switches:

Virtual switch type	Description
External	Provides external access outside of the host machine by mapping to a network adapter in the host, which is used for communication.
Internal	Used to communicate between the virtual machines on a host server and to communicate between the virtual machines and the host itself
Private	Used to only communicate between virtual machines on a Hyper-V host, but does not allow communication with the host itself

Demo virtual switch

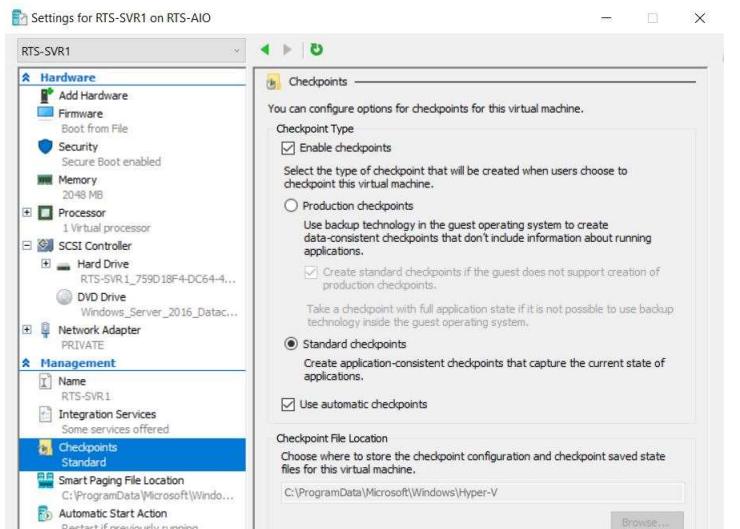
Manage VM states and checkpoints

- A VM can be in one of the following states:
 - Off
 - Starting
 - Running
 - Paused
 - Saved



Manage VM states and checkpoints

- Checkpoints:
 - Allows you to take a snapshot of a virtual machine at a specific point in time
 - Two types of checkpoints
 - Production checkpoints
 - Standard checkpoints
 - Maximum of 50 checkpoints per virtual machine allowed



demo

checkpoint

Lesson 3 overview

This lesson covers virtual machine (VM) migration and available migration options.

Topics:

- High availability options for Hyper-V VMs
- Overview of live migration
- Live migration requirements
- Replicas

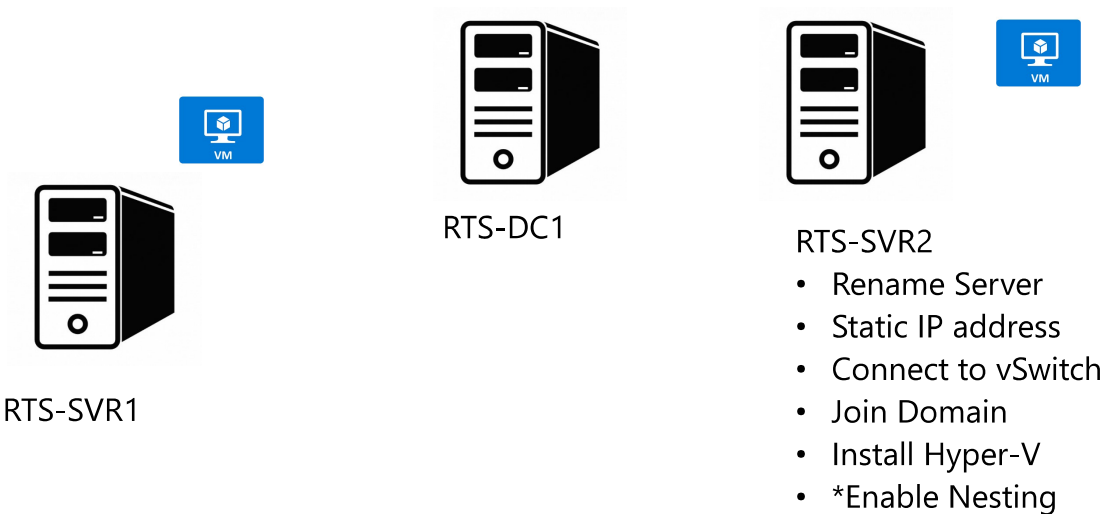
Overview of live migration

- Live migration is a Hyper-V feature in Windows Server
- Used to move running VMs from one Hyper-V host to another without downtime
- Live migration can be performed using the following methods:
 - The Failover Cluster Management console
 - The Virtual Machine Manager Administrator console
 - Hyper-V Manager
 - Windows Management Instrumentation (WMI)
 - Windows PowerShell

VM Import Types

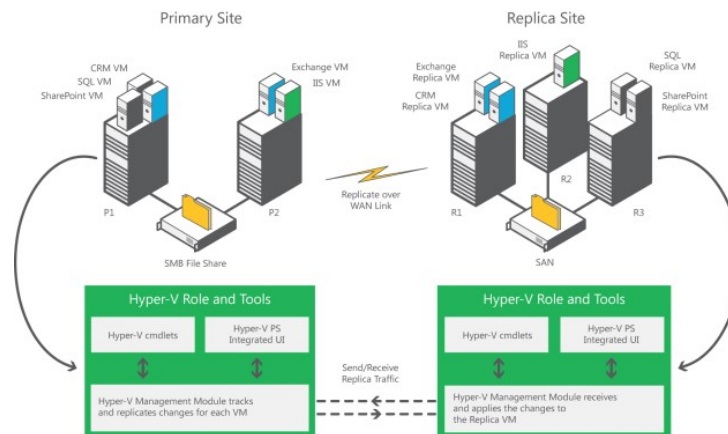
Import Type	Description
Register the virtual machine in-place	The current location of the Export files is where the virtual machine will be run from. When imported, the virtual machine has the same ID as it did at the time of export. The export files become the running state files and cannot be removed.
Restore the virtual machine	You are given an option to store the VM files in a specific location or use the locations default to Hyper-V. This import type creates a copy of the exported file and moves them to the selected location. When imported, the virtual machine has the same ID as it did at the time of export. When the import has completed the exported files remain untouched and can be removed and / or imported again.
Copy the virtual machine	This import type is similar to the Restore type in that you select a location for the VM files. The difference is that when imported the virtual machine has a new unique ID. This allows for the Virtual Machine to be imported into the same host multiple time.

Live Migration



Hyper-V Replica

A Hyper-V Replica has a built-in replication mechanism for virtual machines. A Hyper-V Replica can asynchronously replicate a virtual machine in a primary site to a replica virtual machine in a secondary site, enabling workload continuity and recovery when outages occur.



Overview of Hyper-V Replica

You can establish Hyper-V Replica with a Hyper-V host regardless of its location and domain membership, providing you have network connectivity with it

You also can implement Hyper-V Replica when Hyper-V hosts are members of untrusted domains, if you configure certificate-based authentication

Hyper-V creates a local security group named Hyper-V Administrators

When you limit replication to specific servers, you need to specify a trust group, which is used to identify the servers within which a virtual machine can move

The replica Hyper-V host can authenticate a primary Hyper-V host by using Kerberos authentication or certificate authentication

You should configure Windows Defender Firewall to allow either HTTP or HTTPS Hyper-V Replica traffic

Demo

replica



Module 6: Managing SAN Storage and Failover Clustering

Module overview

Lessons:

- Lesson 1: ISCSI SAN
- Lesson 2: Planning for failover clustering implementation
- Lesson 3: Creating and configuring failover clusters

Lesson 1 overview

Topics:

- ISCSI SAN
- Demo: ISCSI installation and configuration

ISCSI Storage Area Networks (SAN)



RTS-DC1
192.168.1.250
ISCSI Target Server



RTS-SVR1
192.168.1.251
Access Server



RTS-SVR2
192.168.1.253
Access Server

Lesson 2 overview

Topics:

- What is failover clustering?
- High availability with failover clustering
- Failover clustering components
- Cluster quorum in Windows Server
- Considerations for planning failover clustering

What is failover clustering?

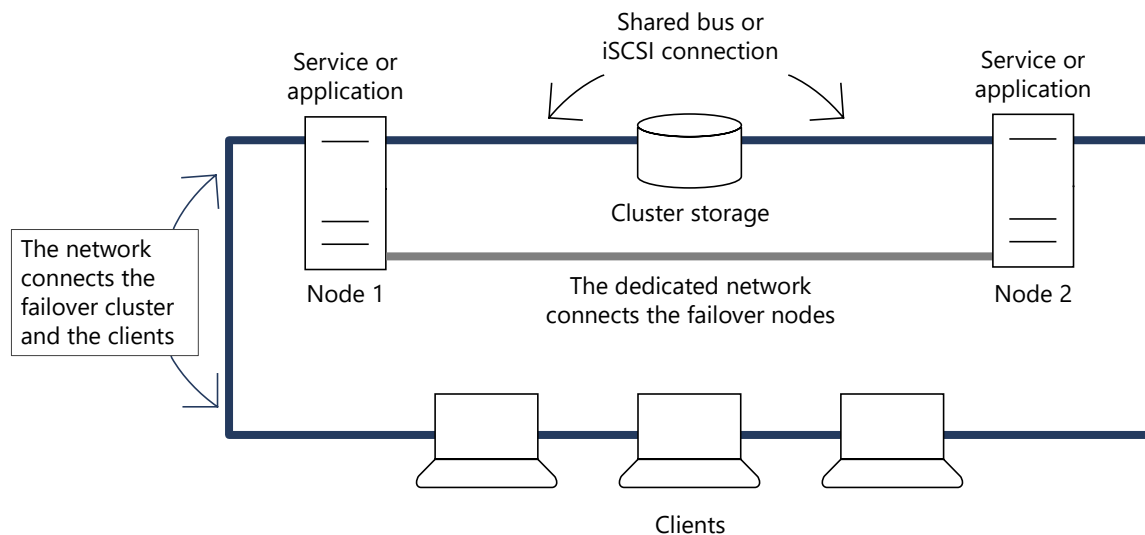
- *Failover clustering* is a group of computers that work together to increase the availability and scalability of clustered roles
- The clustered servers (called *nodes*) are connected by physical cables and by software
- If one or more of the cluster nodes fail, other nodes begin to provide service in a process known as *failover*
- Clustered roles are proactively monitored to verify that they are working properly
- If they are not working another node in the cluster runs the workload

High availability with failover clustering

Availability is a level of service expressed as a percentage of time

- *Highly available* services or systems are available more than 99 percent of the time
- Planned outages typically are not included when calculating availability

Failover clustering components



Cluster quorum in Windows Server

In failover clusters, quorum defines the consensus that enough cluster members are available to provide services.

Quorum:

- Is based on votes in Windows Server
- Enables nodes, file shares, or a shared disk to have a vote, depending on the quorum mode
- Enables the failover cluster to remain online when sufficient votes are available



Configure quorum options

Use dynamic quorum mode with:

- A disk witness
- A file share witness
- The Azure Cloud Witness

Use all other quorum modes only in specific use cases

The default and recommended best practice is to always use dynamic quorum

Lesson 3 overview

Topics:

- The Validation a Configuration Wizard and cluster support policy requirements
- Create a failover cluster
- Configure storage
- Configure networking
- Configure quorum options
- Configure roles
- Manage failover clusters
- Configure cluster properties

The Validate a Configuration Wizard and a cluster support policy requirements

The Validate a Configuration Wizard is used to perform a variety of tests to ensure the cluster components are configured in a supportable manner.

Before creating a new failover cluster, confirm the configuration to ensure all validation tests are passed.

Cluster validation is intended to:

- Ensure clustering is working properly
- Find hardware or configuration issues
- Perform diagnostic tests
- Ensure requirements for:
 - Hardware
 - Network/Infrastructure
 - Software

Create a failover cluster

To create a failover cluster, you'll need to:

- Verify the prerequisites
- Install the Failover Clustering feature on each node
- Run the Validate a Configuration Wizard
- Create the cluster using:
 - The Create Cluster Wizard, or
 - Windows Admin Center
- Create clustered roles

Configure storage

Failover clusters require shared storage to provide consistent data to a virtual server after a failover

Shared storage options include:

- SAS
- iSCSI
- Fibre Channel
- Shared **.vhdx**

Clustered storage spaces can also be implemented to achieve high availability at the storage level

Configure networking

To configure networking:

- The network hardware must be compatible with Windows Server
- In the network infrastructure that connects your cluster nodes, avoid having single points of failure

Configure roles

To configure roles:

1. Install the Failover Clustering feature
2. Verify the configuration
3. Create a cluster
4. Install the role on all cluster nodes by using Server Manager
5. Create a clustered application by using the Failover Clustering Management snap-in
6. Configure the application
7. Test the failover

Manage failover clusters

To manage failover clusters:

- Add nodes after you create a cluster
- Pause nodes, which prevent resources from running on that node
- Evict nodes from a cluster, which removes the node from the cluster configuration

These actions are available in the **Failover Cluster Management Console**, in the **Actions** pane

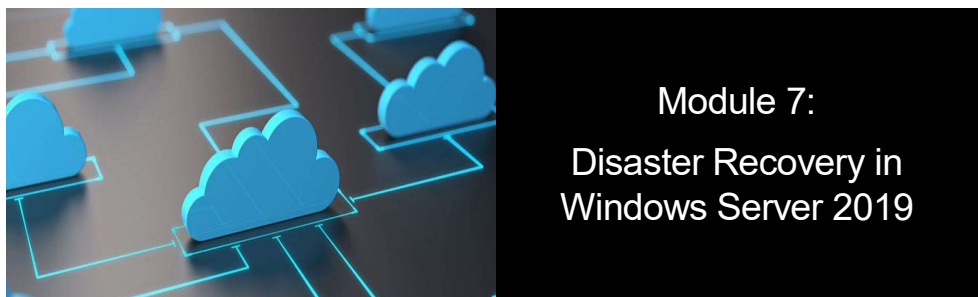
Configure failover and failback

To control how the cluster responds, adjust the failover and failback settings.

Include preferred owners

Considerations for using preferred owners:

- Set preferred owners are set on the clustered role
- Set multiple preferred owners can be set in an ordered list
- Setting preferred owners gives control over:
 - The order in which a role selects a node to run
 - The roles that can be run on the same nodes
- Options to modify failover and failback settings:
 - Setting the number of times the Cluster service restarts a clustered role in a set period
 - Setting or preventing failback of the clustered role to the preferred node when it becomes available



Module overview

Lessons:

- Lesson 1: Back up and restore infrastructure in Windows Server

Lesson 1 overview

Topics:

- Overview of Windows Server Backup
- Implement backup and restore
- Back up and restore Hyper-V VMs
- Overview of Azure Backup
- Implement backup and restore with Azure Backup

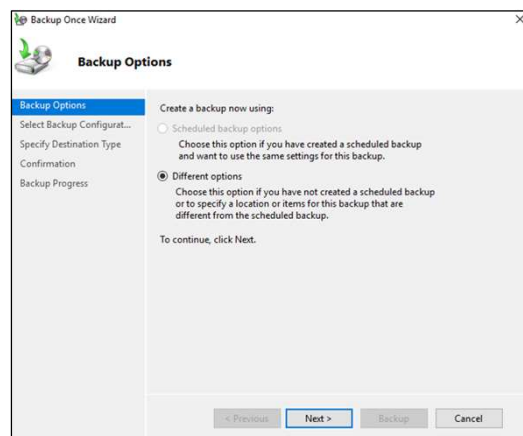
Overview of Windows Server Backup

Windows Server Backup provides you with the ability to perform backup and recovery in a Windows Server environment

By using Windows Server Backup you can backup:

- A full server (all volumes), or just selected volumes
- Individual files and folders
- System state
- Individual virtual machines on a Hyper-V host

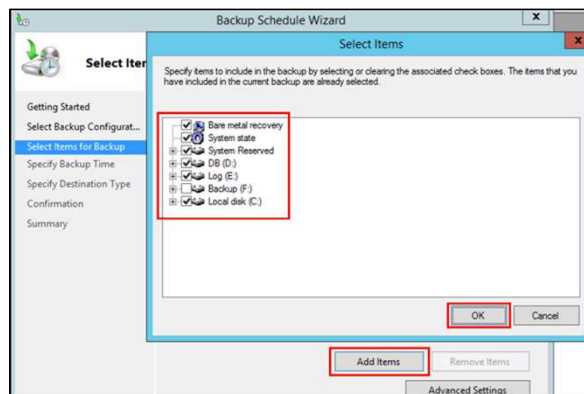
WBAAdmin is a command-line utility built into Windows Server



Implement backup and restore

Depending on what you need to backup, the procedures and options in Windows Server Backup might vary
Some of the most common backup procedures that you should consider include:

- Backing up file servers and web servers
- Backing up AD DS
- Backing up Microsoft Exchange Server



Back up and restore Hyper-V VMs

You can use the following methods to back up VMs:

- Backup the VM from the host
- Backup the VM's VHDs
- Backup inside the VM

You can perform online backups that do not incur VM downtime, if you meet the following conditions:

- The VM being backed up has integration services installed and enabled
- Each disk that the VM uses is running NTFS file system basic disks
- The VSS is enabled on all volumes within the VM

Overview of Azure Backup

Azure Backup is a subscription service that you can use to provide off-site protection against critical data loss caused by disasters

Azure Backup replaces or extends your existing on-premises or off-site backup solution

Some of the most important features in Azure Backup include:

- Automatic storage management
- Unlimited scaling
- Data encryption
- Offload on-premises backup
- Back up Azure VMs

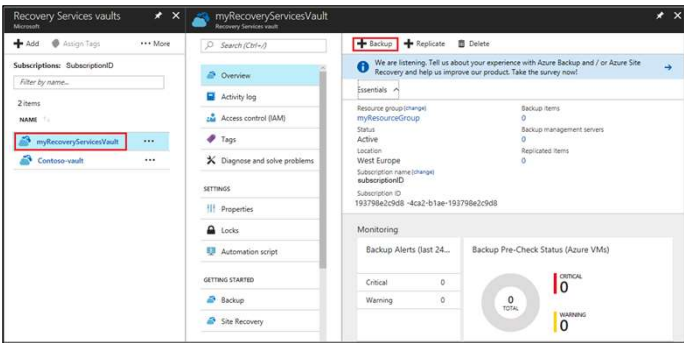
Implement backup and restore with Azure Backup

To use Azure Backup, you need to install a backup agent on your local servers, and you need to configure an Azure Recovery Services vault

You can use Recovery Services vaults to hold backup data for various Azure services such as VMs (Linux or Windows) and Azure SQL databases

Within an Azure subscription, you can create up to 25 Recovery Services vaults per region

Azure Backup for files and folders relies on the Azure Recovery Services agent to be installed on the Windows client or server



Module 8: Implementing WSUS on Windows Server 2019

Module Overview

This module describes how ensure that your servers and clients are current with Windows updates

- Lessons:
 - Lesson 1: Windows Server update management

Lesson 1 overview

This lesson describes WSUS. It provides infrastructure to download, test, and approve updates which help block attacks

- Topics:
 - Overview of Windows Update
 - What is WSUS?
 - WSUS server deployment options
 - The WSUS update management process

Overview of Windows Update

- Windows Update is a Microsoft service that provides updates for Microsoft software
- Orchestrator on devices scans for and downloads updates
- Clients and servers can be configured to get updates from the Windows Update Services server

What is WSUS?

WSUS provides an infrastructure for managing updates for Windows devices

WSUS allows you to:

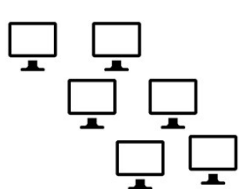
- Choose the updates you want to download
- Test updates before broad deployment
- Choose which devices get updates and when they receive them
- Track status of updates



WSUS Requirements

Prerequisites:

- 1.4 gigahertz (GHz) or faster x64 processor
- 2 gigabytes (GB) of random-access memory (RAM) or greater (above that needed for other roles)
- 10 GB or greater
- 100 megabits per second (Mbps) or greater network adapter
- .NET Framework 4.0
- Microsoft Report Viewer Runtime 2012
- Windows Internal Database or Microsoft SQL Server



Microsoft
Update

WSUS server deployment options

WSUS implementation:

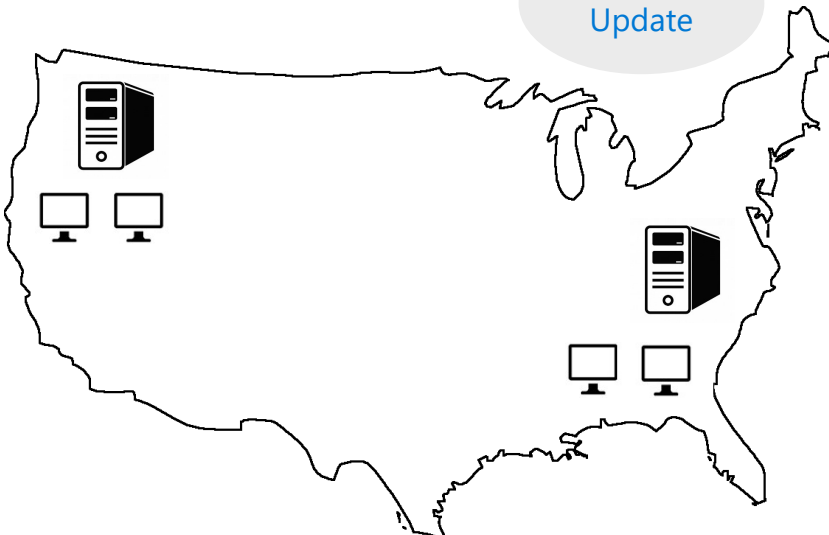
- Single server
- Multiple servers

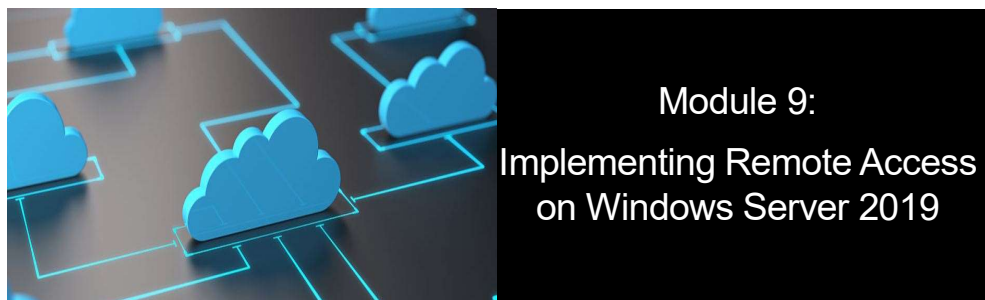
WSUS hierarchies:

- Autonomous mode
- Replica mode

WSUS database:

- Windows Internal Database
- SQL Server database



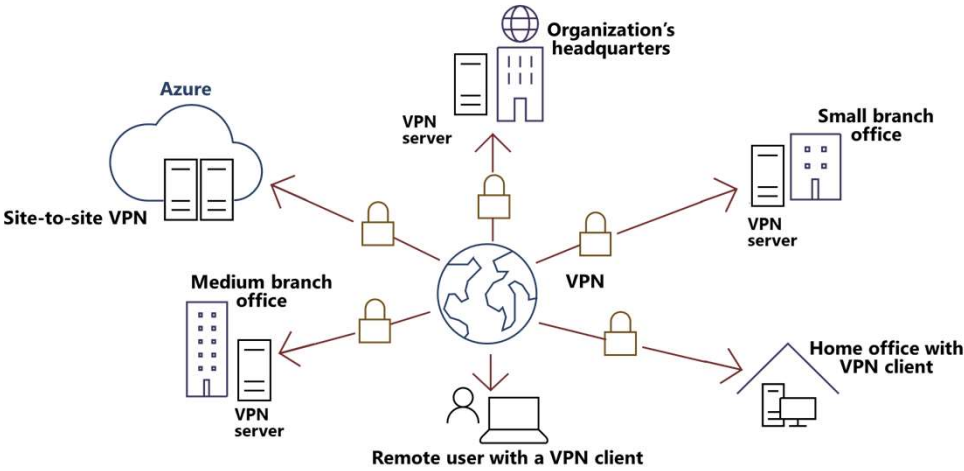


Lesson 1 overview

- Topics:
 - VPN scenarios
 - Options for VPN tunneling protocols
 - VPN authentication options
 - Configure a VPN by using the Getting Started Wizard

VPN Scenarios

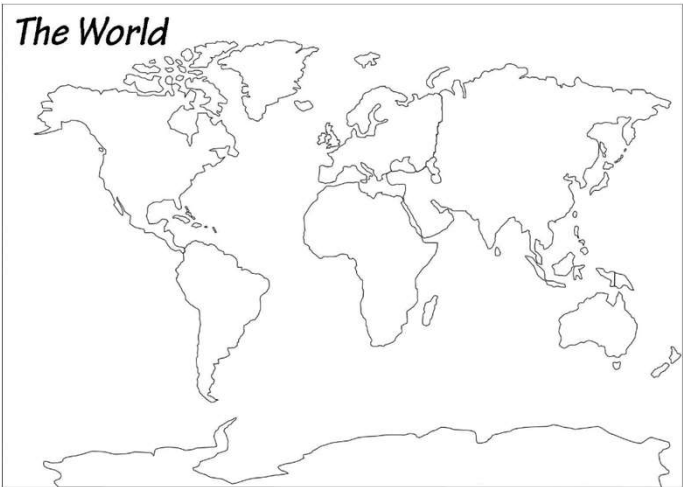
A VPN provides a point-to-point connection between a private network's components by using a public network, such as the Internet.



Options for VPN tunneling protocols

Windows Server supports four VPN tunneling protocols.

Tunneling protocol	Firewall access
PPTP	TCP port 1723
L2TP/IPsec	UDP port 500, UDP port 4500 and UDP port 1701
SSTP	TCP port 443
IKEv2	UDP port 500



VPN authentication options

Protocol	Description	Security level
PAP	Uses plaintext passwords. Typically used if the remote access client and remote access server cannot negotiate a more secure form of validation.	The least secure authentication protocol. Does not protect against replay attacks, remote client impersonation, or remote server impersonation.
CHAP	A challenge-response authentication protocol that uses the industry-standard MD5 hashing scheme.	An improvement over PAP in that the password is not sent over the PPP link. Requires a plaintext version of the password to validate the challenge response. Does not protect against remote server impersonation.
MS-CHAPv2	An upgrade of MS-CHAP. Provides two-way authentication, also known as mutual authentication. The remote access client receives verification that the remote access server to which it is dialing in to has access to the user's password.	Provides stronger security than CHAP.
EAP	Allows for arbitrary authentication of a remote access connection through the use of authentication schemes, known as EAP types.	Offers the strongest security by providing the most flexibility in authentication variations.



Module 10: Performance Monitoring in Windows Server 2019

Module Overview

Overview of Windows Server monitoring tools

Using Performance Monitor

Monitoring event logs for troubleshooting

Lesson 1: Overview

Overview of Task Manager

Overview of Resource Monitor

Overview of Performance Monitor

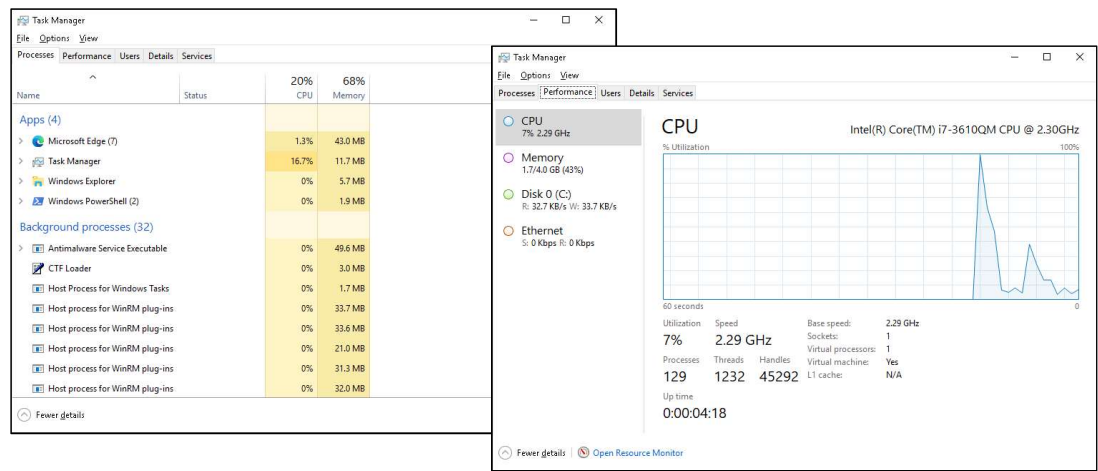
Overview of Reliability Monitor

Overview of Event Viewer

Monitoring a server with Windows Admin Center

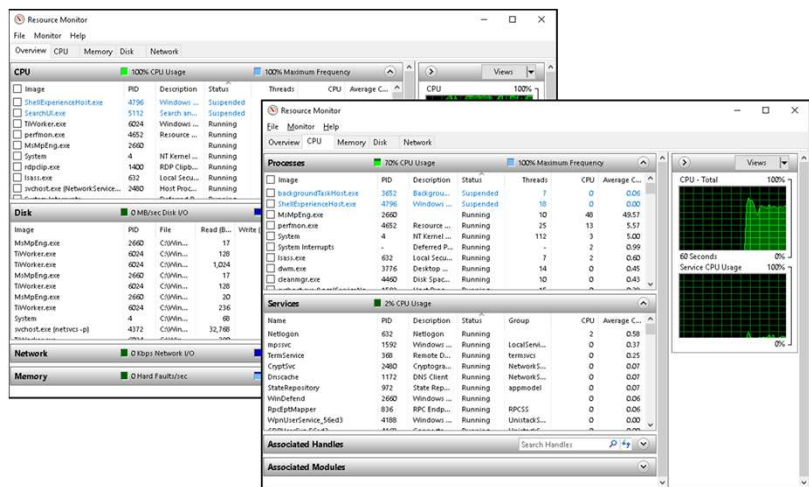
Overview of Task Manager

Task Manager helps you to identify and resolve performance-related issues



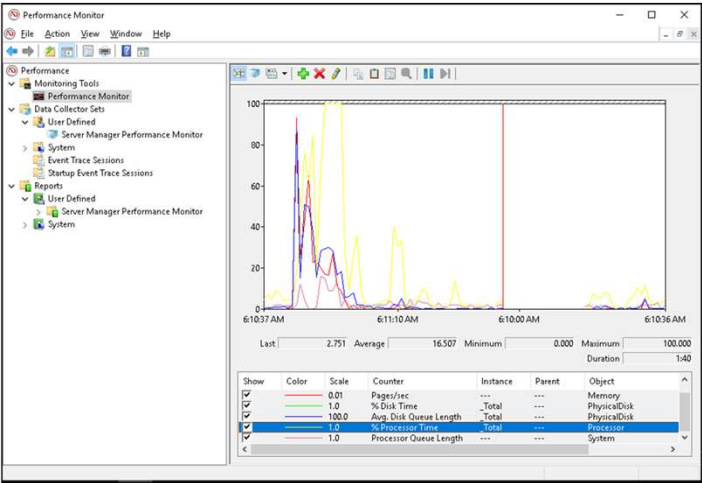
Overview of Resource Monitor

Resource Monitor provides an in-depth understanding at the real-time performance of your server



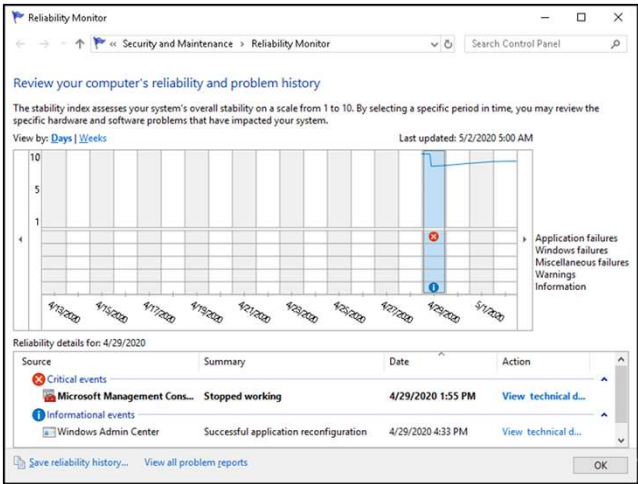
Overview of Performance Monitor

Performance Monitor enables you to observe current performance statistics or to study historical data that Data Collector Sets have gathered



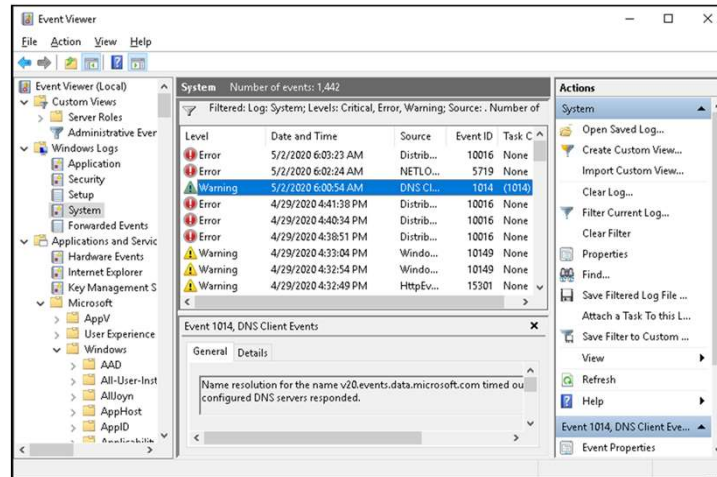
Overview of Reliability Monitor

Reliability Monitor monitors hardware and software issues that occur during the selected time interval and assigns a number called the stability index that indicates the server's reliability



Overview of Event Viewer

Event Viewer provides categorized lists of essential Windows log events and log groupings for individual installed applications and specific Windows component categories



Overview of baselines, trends, and capacity planning

By calculating performance baselines for your server environment, you can more accurately interpret real-time monitoring information

By establishing a baseline, you can:

- Interpret performance trends
- Perform capacity planning
- Identify bottlenecks
- Analyze performance trends to predict when existing capacity is likely to be exhausted
- Plan the capacity for the key hardware components such as processor, disk, memory, and network

Considerations for monitoring virtual machines

Considerations for monitoring virtual machines:

- Virtual machines must be assigned sufficient resources for their workload
- If multiple virtual machines run on a host, ensure the host has enough resources
- Resources are shared, so performance of one virtual machine can affect the performance of others
- You must remember to monitor the resource utilization on the host as well as the guests

Thanks for attending

Follow-on Courses:

AZ-900: Azure Fundamentals

YouTube Channel